

Cirrhosis & Ascites

Prof. Wai-Kay Seto

MBBS MD FRCP

FHKCP FHKAM AGAF

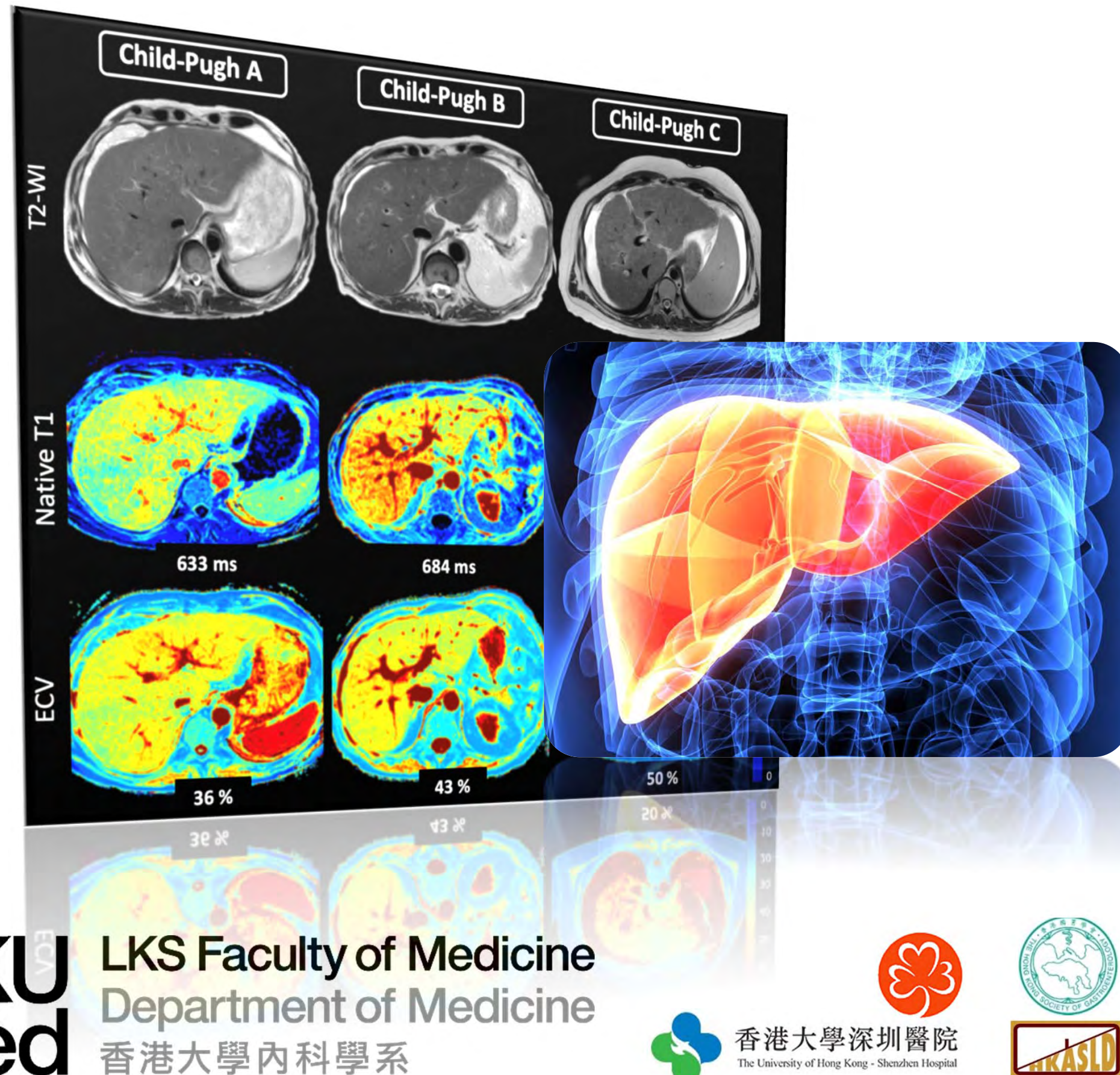
Clinical Professor

The University of Hong Kong



**HKU
Med**

LKS Faculty of Medicine
Department of Medicine
香港大學內科學系



WCS Outline

- Cirrhosis: aetiology and terminology
- Portal Hypertension: pathogenesis
 - Variceal disease
- Ascites: pathogenesis
 - Ascites
 - Spontaneous bacterial peritonitis
 - Hepatic hydrothorax
 - Portal vein thrombosis
- Acute Kidney Injury / Hepatorenal syndrome

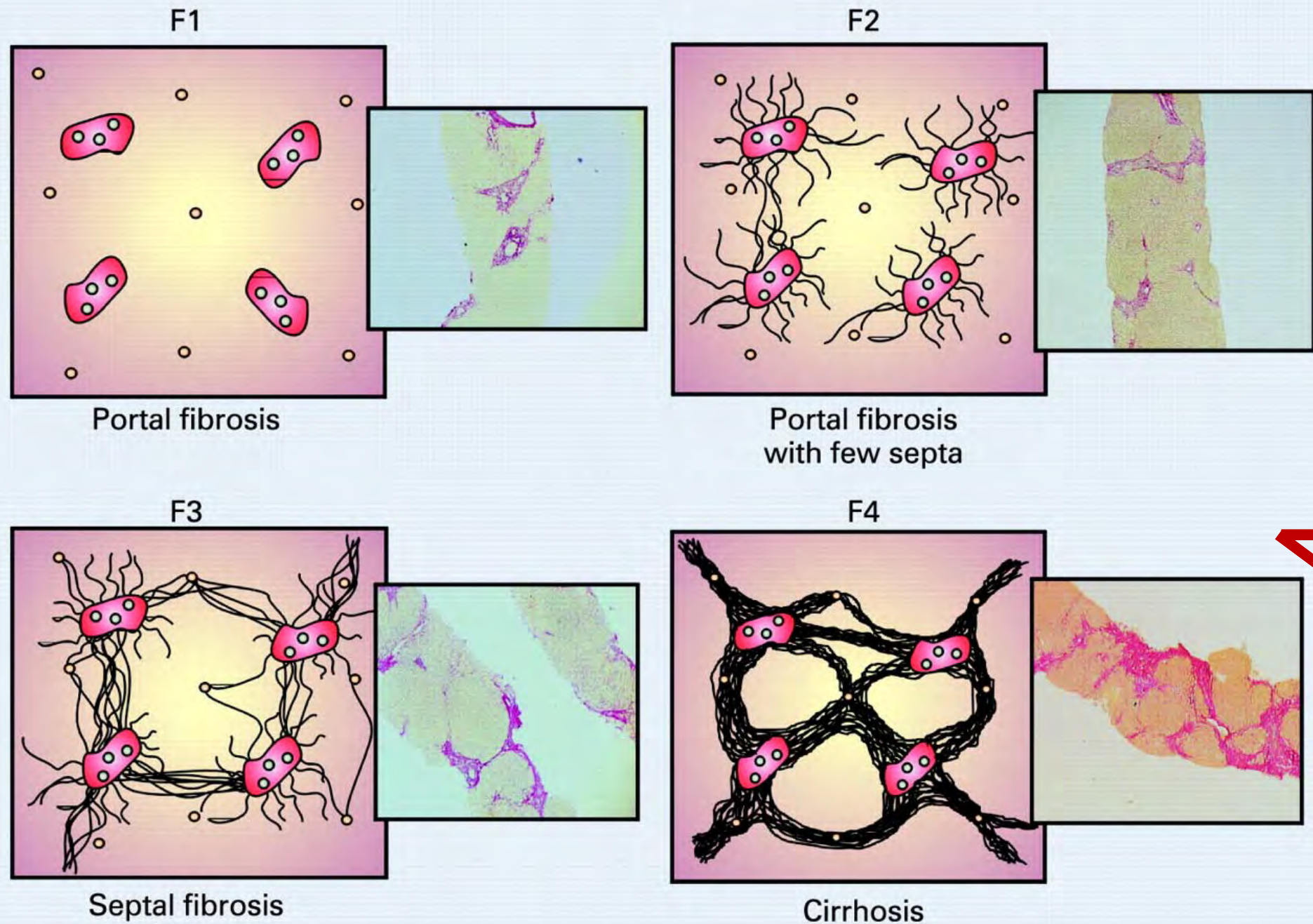
Covered Elsewhere:

- Liver failure
- Hepatic encephalopathy
- Hepatocellular carcinoma
- Liver transplant

Cirrhosis & Decompensation

Cirrhosis

Late stage of liver fibrosis with distortion of liver architecture & formation of regenerative nodules



**Compensated
Child's A**

**Decompensated
Child's B/C**

Etiology of Cirrhosis: Prevalence

WHO estimates

Cirrhosis	Global	Mainland China	Hong Kong
HBV	42%	68%	64%
HCV	21%	7%	10%
Heavy alcohol use prevalence	Variable	11%	5%
MAFLD prevalence	Variable	5%	18%

Prognostic Factors for Cirrhosis

Child-Pugh Score

	1 point	2 points	3 points
Bilirubin (umol/L)	<34	34-51	>51
Albumin (g/L)	>35	28-35	<28
INR	<1.7	1.7-2.3	>2.3
Ascites	None	Mild	Refractory
Encephalopathy	None	Gr 1-2	Gr 3-4
Child classes:	A	5 - 6 points	
	B	7 - 9 points	
	C	10 - 15 points	

* For bilirubin, 1 mg/dl = 17.1 μmol/L

Is Cirrhosis Reversible?

Treating the underlying disease

HBV

Majority
(but not all)
Long-term
nucleoside analogue
therapy

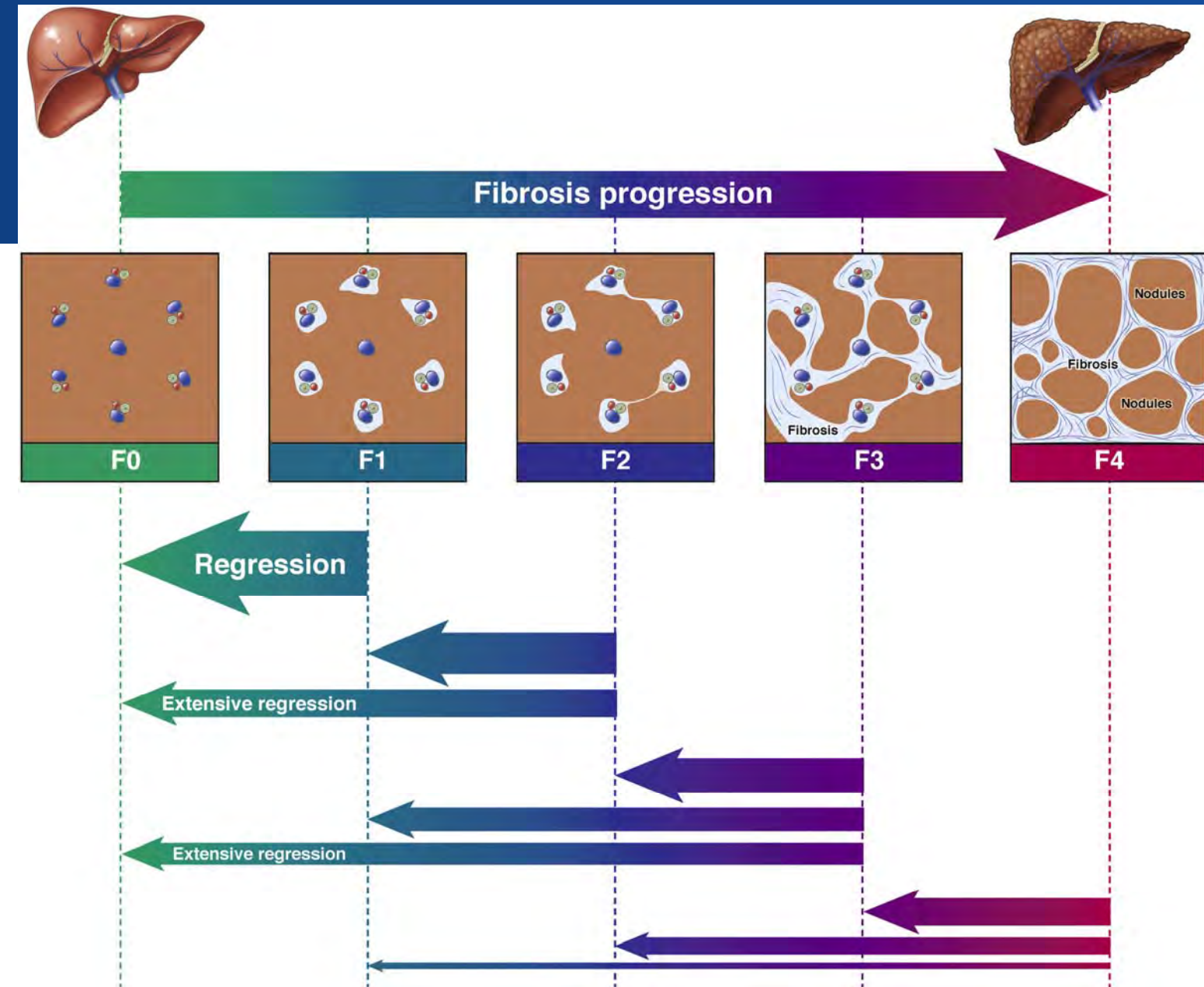
HCV

Some
(but not all)
Direct antiviral therapy
Some degree of
fibrosis regression

Alcohol

Uncertain

Alcohol abstinence
↓ Inflammation
↓ Fat
? Fibrosis



Treating underlying disease

(generally speaking)

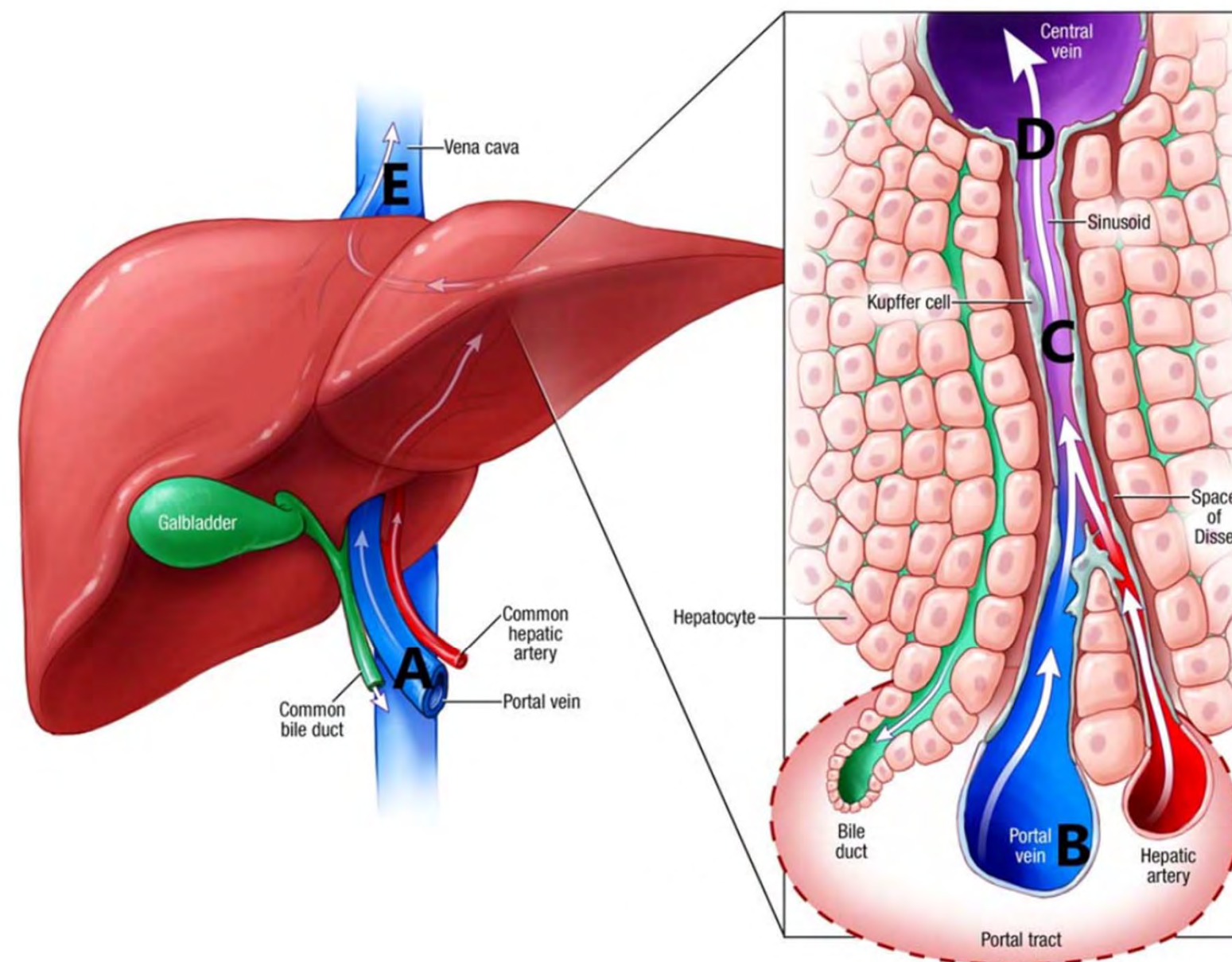
Child's A/B: ↓ complications / hospitalization

Child's C: Less impact

Portal Hypertension

Resistance to portal blood flow

- Pre-hepatic (A)
 - Portal Vein Thrombosis
- Hepatic
 - **Cirrhosis (C)**
 - Other uncommon causes (B, D)
 - *Schistomomiasis / congenital hepatic fibrosis / veno-occlusive disease etc.*
- Post-hepatic (E)
 - *Cardiac failure / Budd-Chiari (hepatic vein thrombosis)*



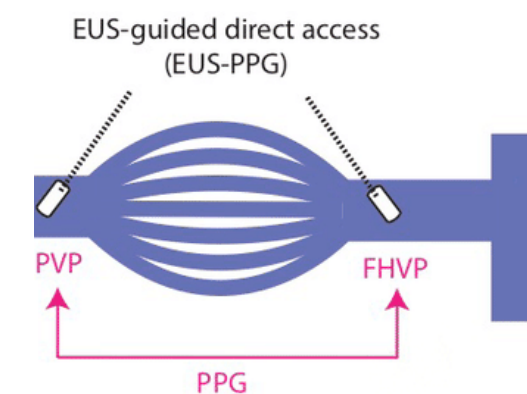
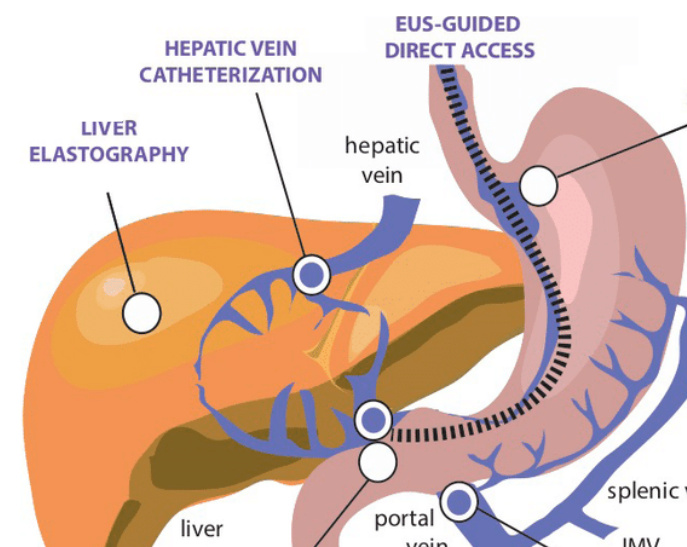
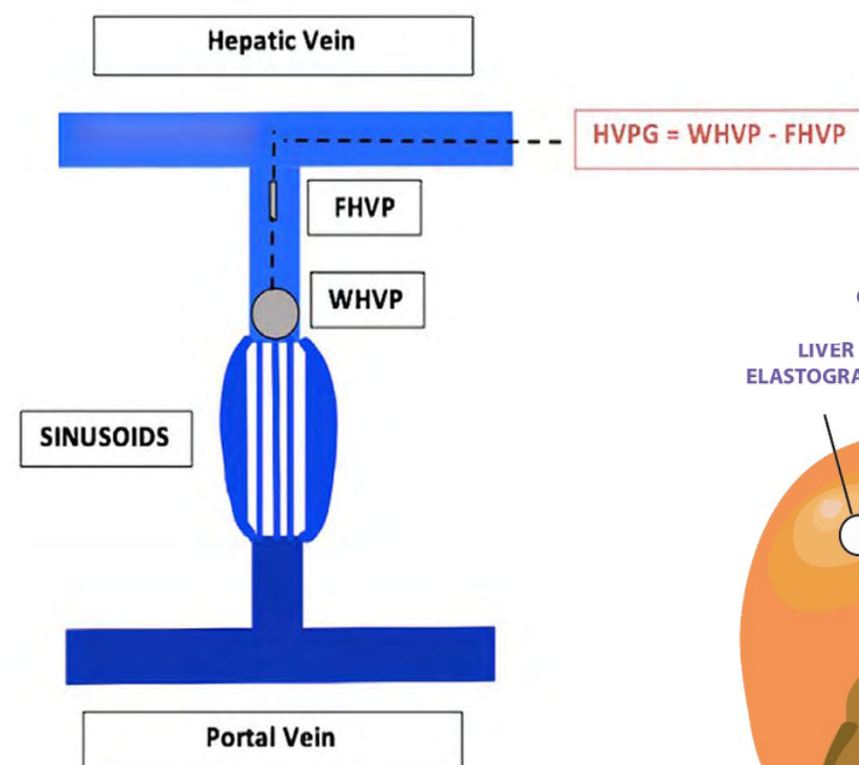
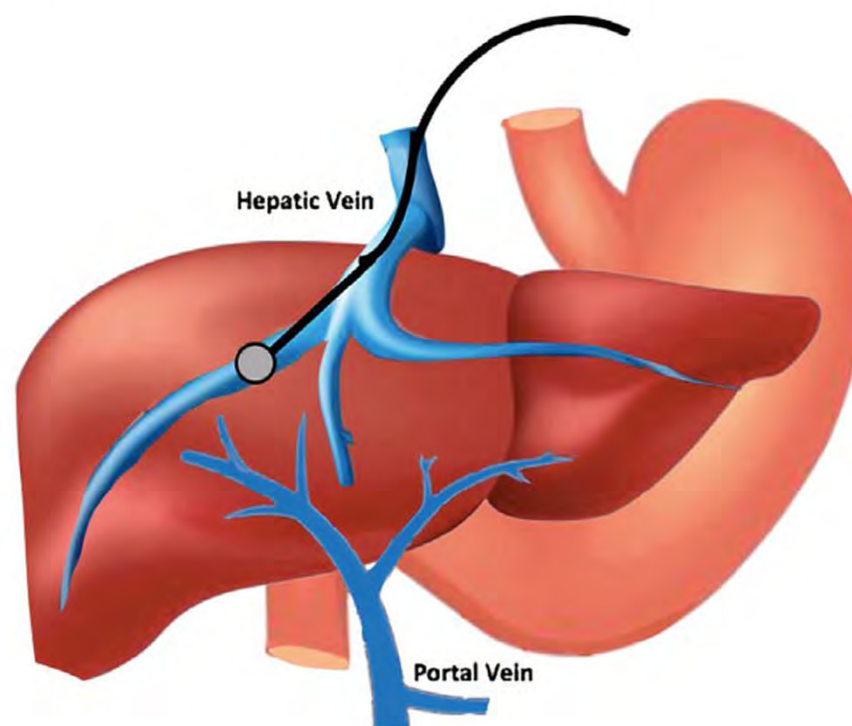
Detection of Portal Hypertension

Before symptoms & complications

EUS-guided portal pressure measurement (not widely performed)

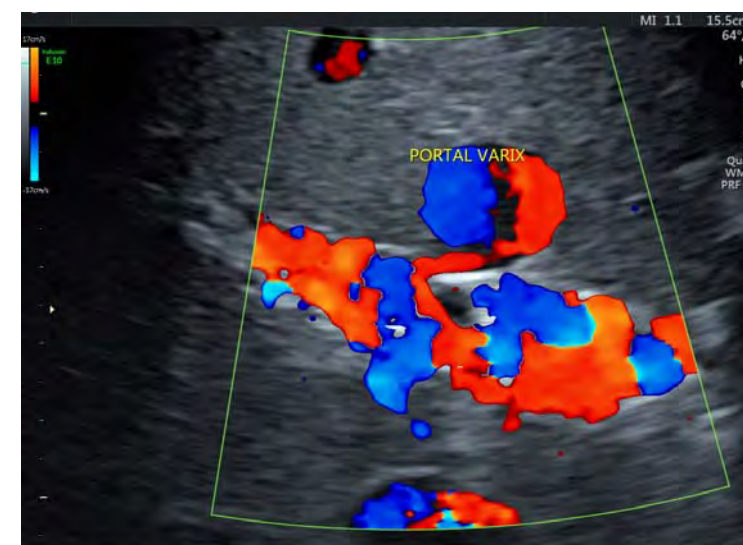


Hepatic Venous Pressure Gradient Measurement



Gold Standard
 $HVPG = WHVP - FHVP^*$
 $HVPG \geq 10$ = clinically significant portal hypertension
 Not widely performed

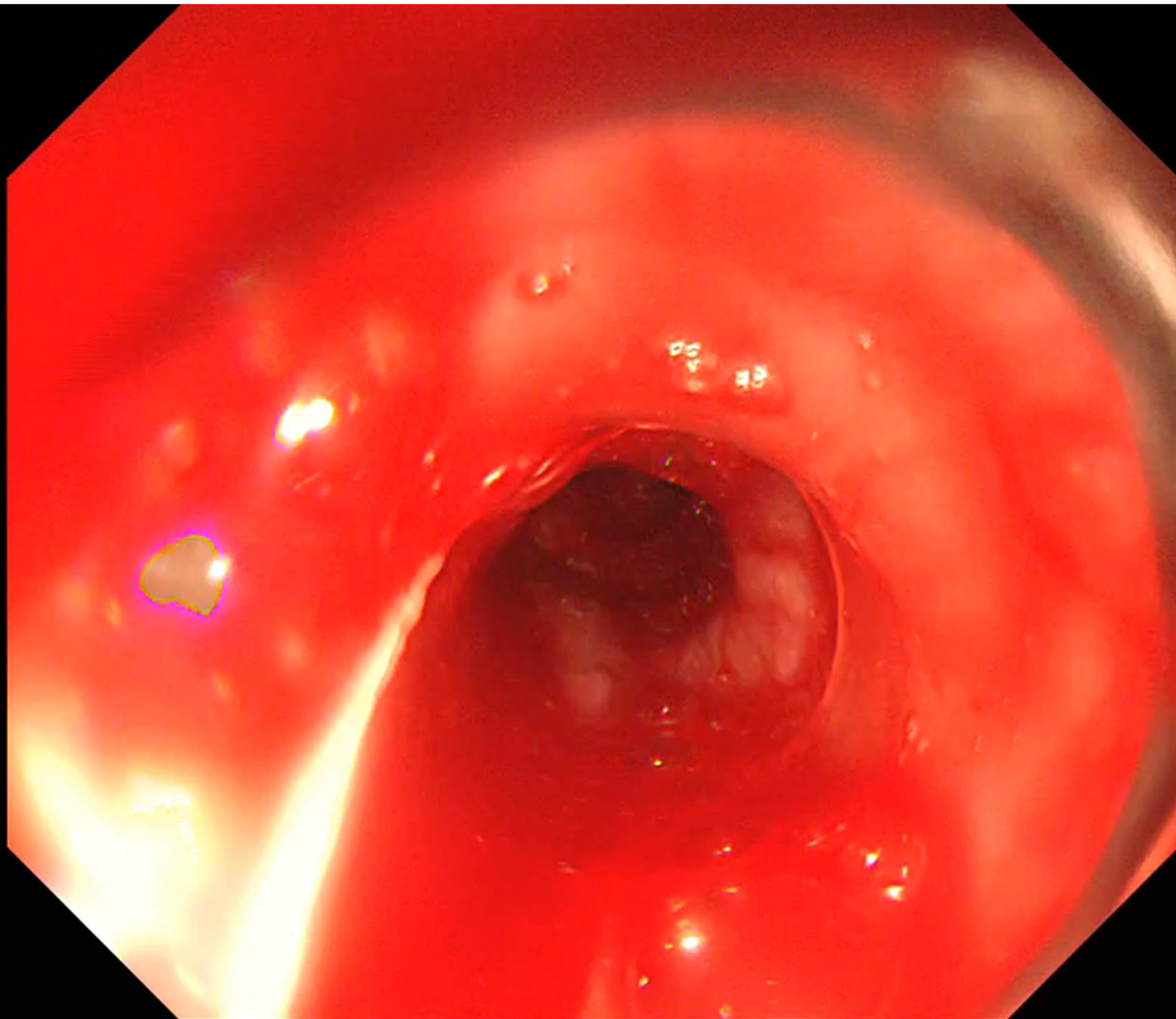
*Hepatic venous pressure gradient / wedged hepatic venous pressure / free hepatic venous pressure



Ultrasound
 (Not sensitive)

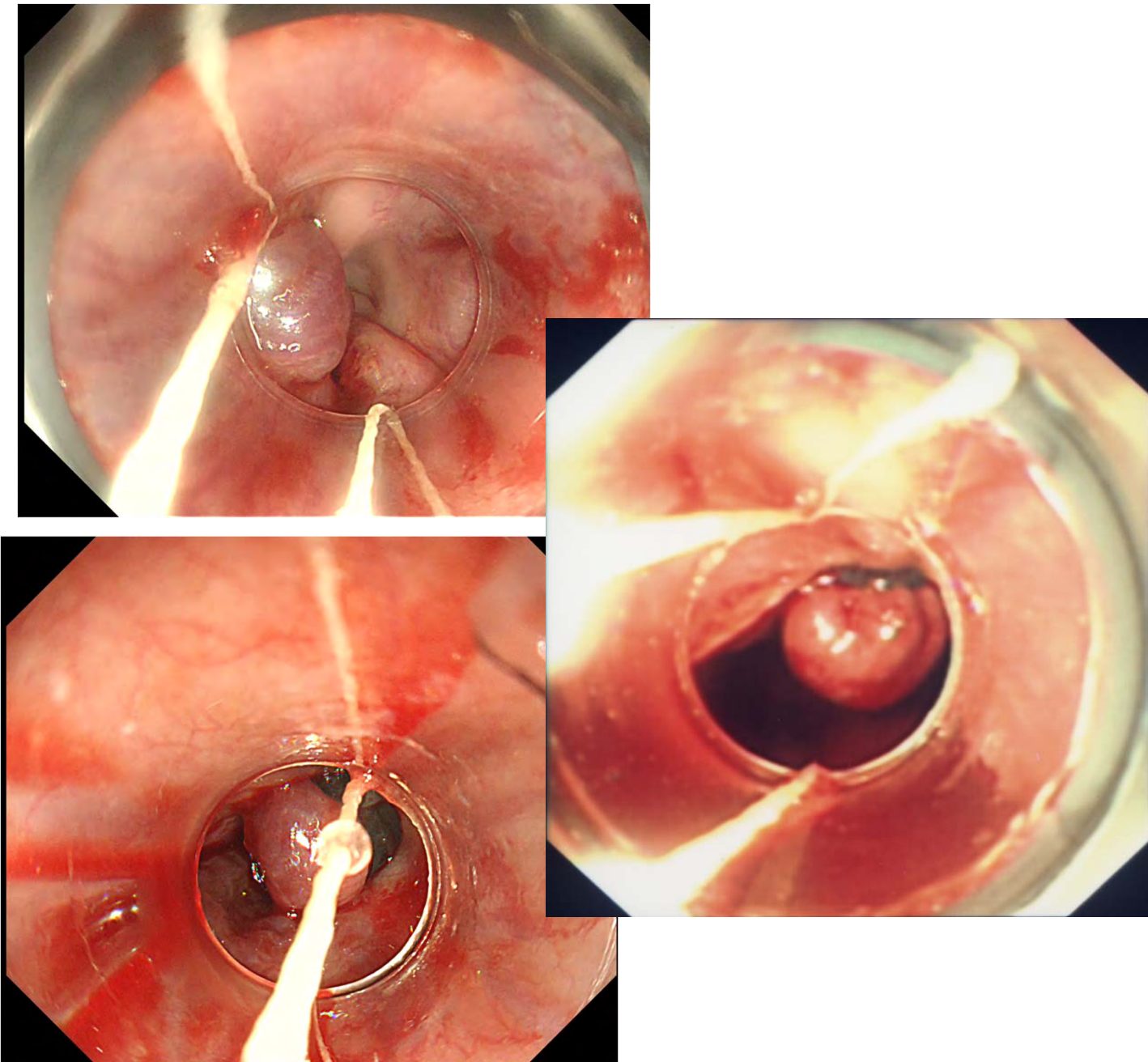
Splenomegaly
 Nodular small liver
 ↑ Portal vein diameter
 ↓ Portal vein flow
 Varices

Acute Esophageal Variceal Bleeding



- HVPG ≥ 12 mmHg
- Present in cirrhosis:
 - Child's A: 40%
 - Child's B/C 70%
- Mortality
 - 15-25%

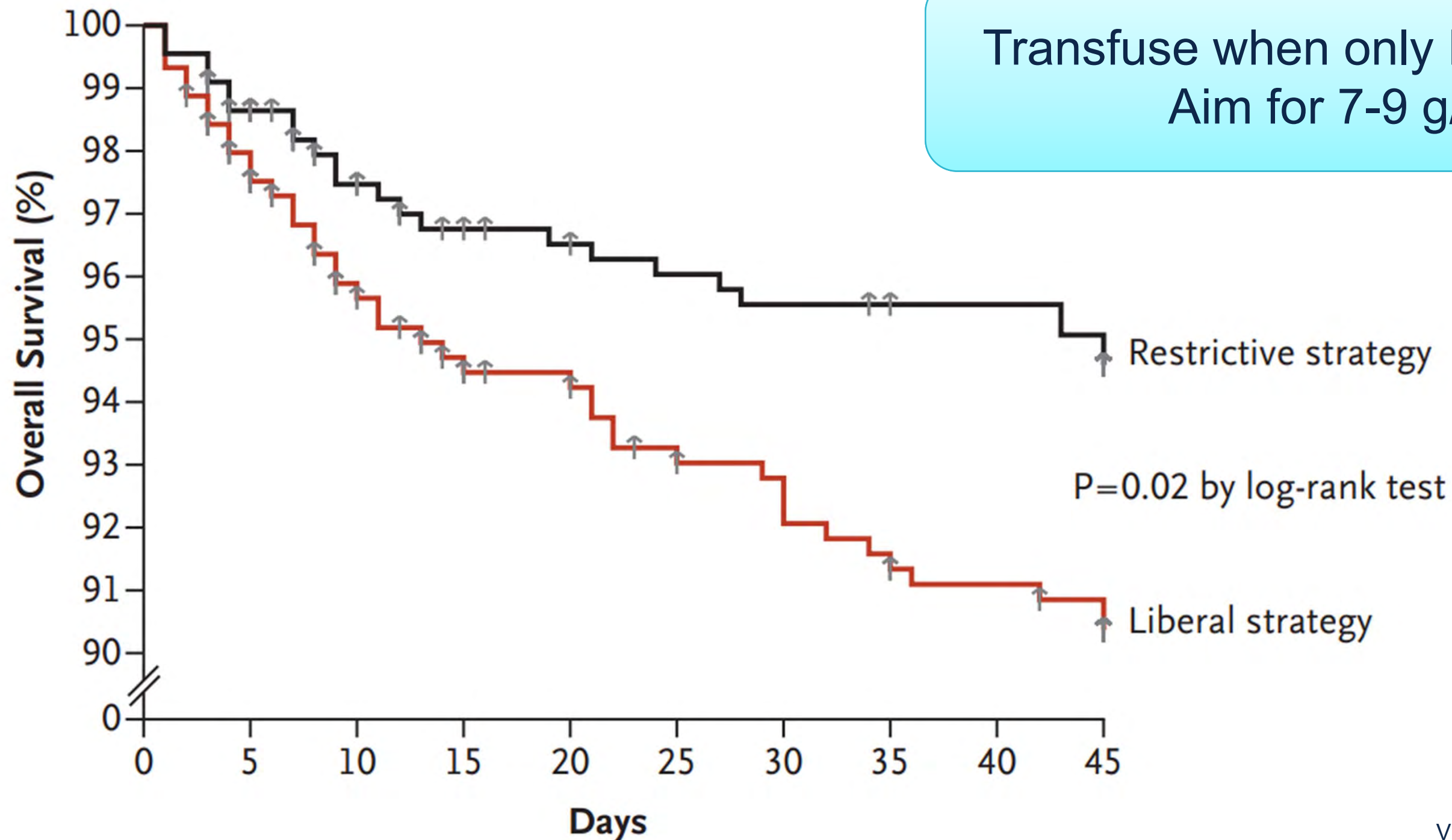
Managing Acute Esophageal Variceal Bleeding



- Pre-endoscopy:
 - As per upper GI bleeding
- Correct coagulopathy / ↓ platelet
- **Restrictive** transfusion strategy
- Antibiotic prophylaxis (cephalosporin)
- Endoscopy:
 - **Band ligation** or
 - Sclerotherapy injection
- Vasoconstrictor
 - **Intravenous terlipressin** or
 - Intravenous somatostatin / octreotide
- Management cirrhotic complication (e.g. HE)

Restrictive Transfusion Strategy

*Especially significant in **liver cirrhosis***



Managing Acute Esophageal Variceal Bleeding

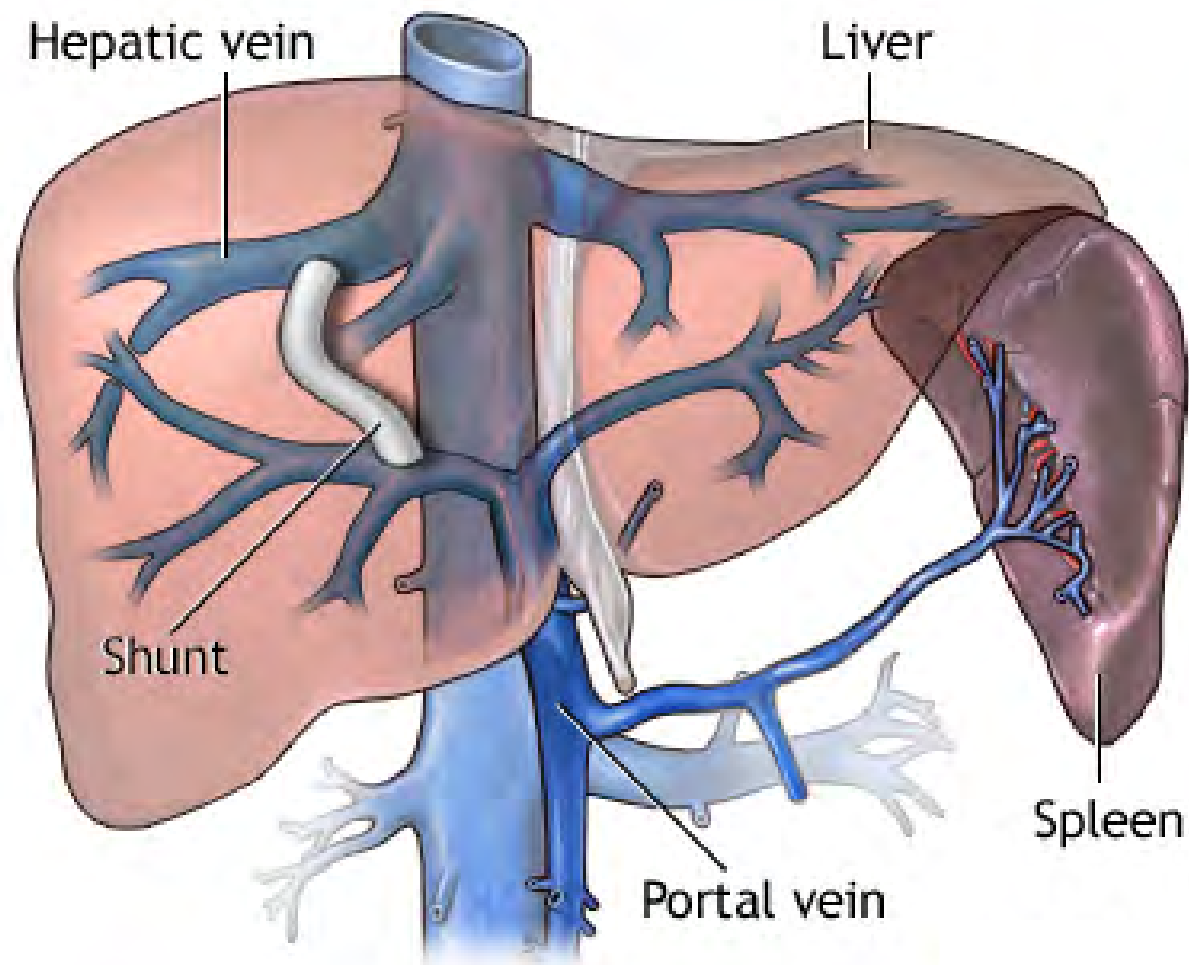
- Band ligation
 - Complications are few:
 - Transient esophageal pain
 - Ulceration (rarely bleed)
 - Sloughing of bands
 - Mediastinitis / infection
 - Reassessment scope a few weeks later
 - To consider further banding
- Drugs
 - **iv terlipressin**
 - Synthetic analog of vasopressin
 - Reduces mortality
 - Side effect: hyponatraemia, ischaemic injury
 - **iv somatostatin**
 - Inhibit release of vasodilator hormones
 - Splanchnic vasoconstriction

Difficult Esophageal Variceal Bleeding

Limited availability in HK

- **TIPSS**

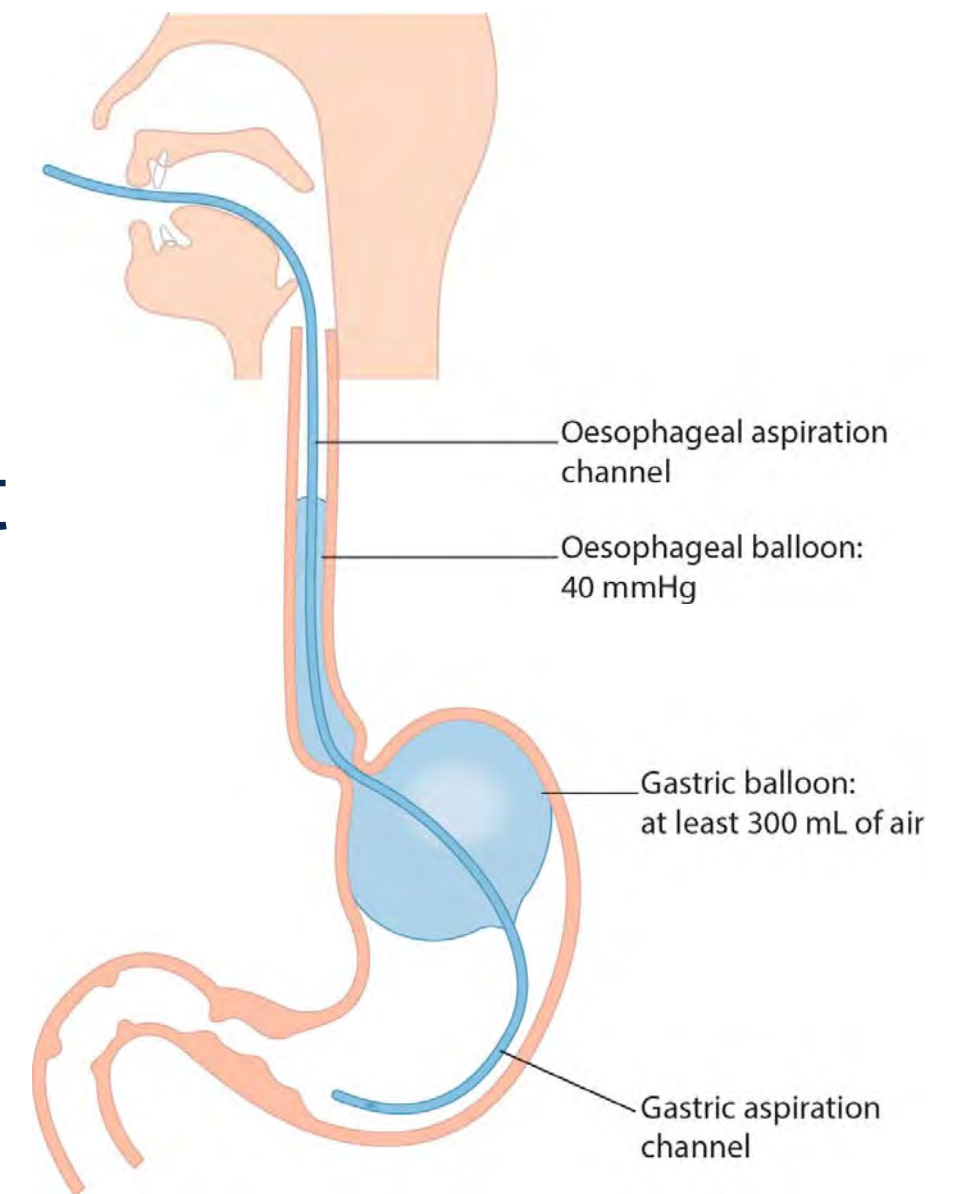
- Transjugular Intrahepatic Portosystemic Shunt



- Other indications:
 - Refractory ascites
 - Refractory hydrothorax
- Contraindications
 - Right heart failure
 - Pulmonary hypertension
 - Hepatic encephalopathy (relative)
- Complications:
 - Acute (infection / bleeding / rupture)
 - HE
 - Stent occlusion / migration

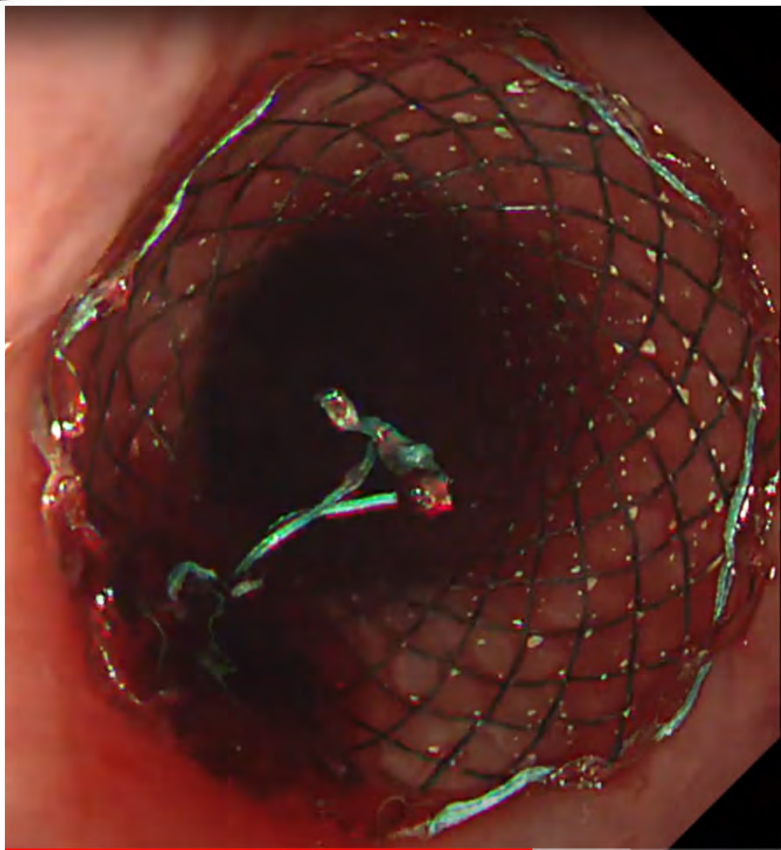
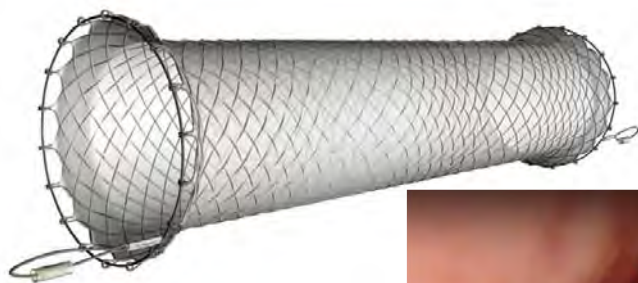
Difficult Esophageal Variceal Bleeding

- Balloon tamponade
 - Sengstaken-Blackmore Tube
 - Intubation to prevent aspiration
 - Temporary measure for definitive treatment
 - Complications:
 - Aspiration
 - Rebleeding
 - Esophageal rupture
 - Try to avoid!

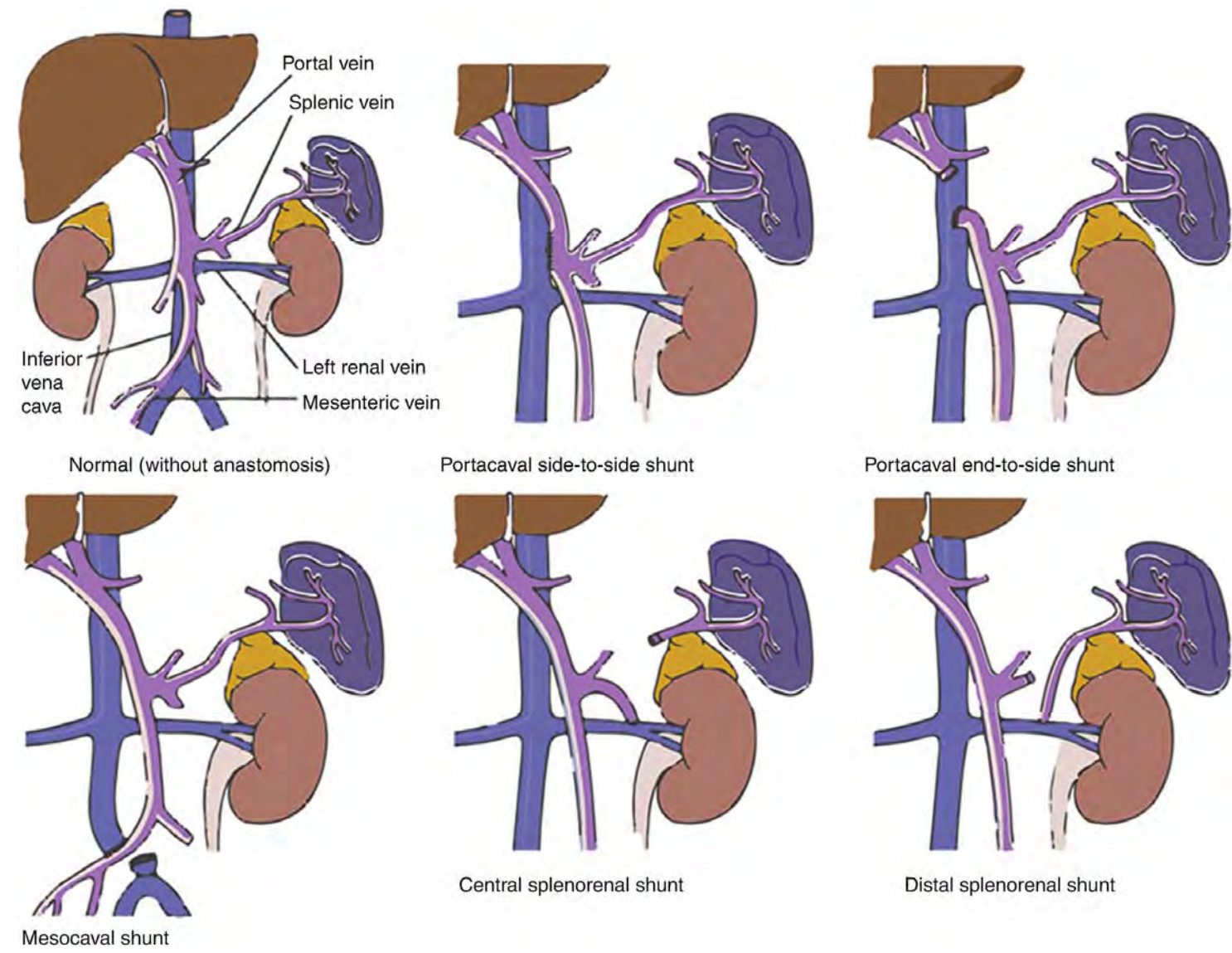


Difficult Esophageal Variceal Bleeding

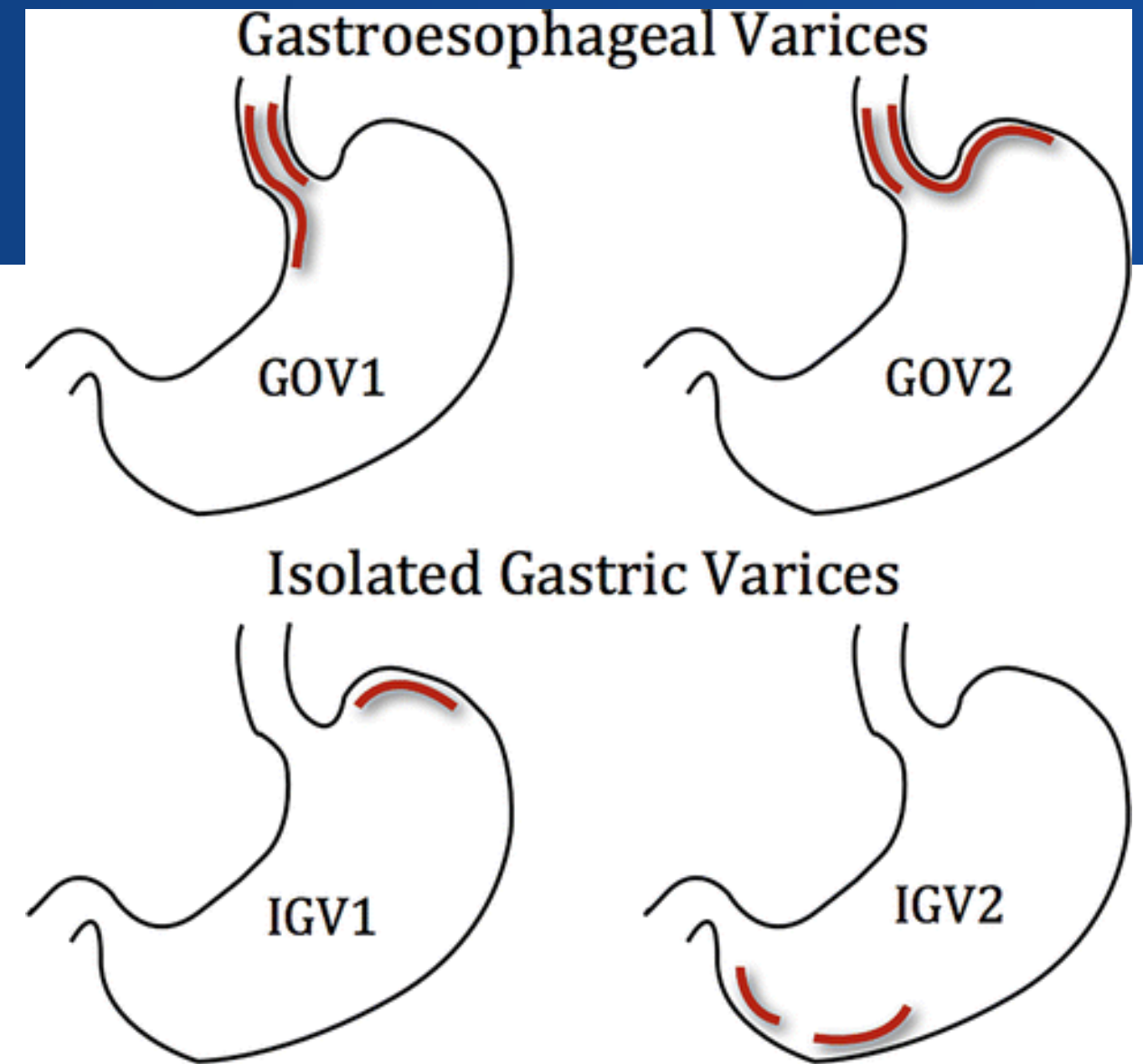
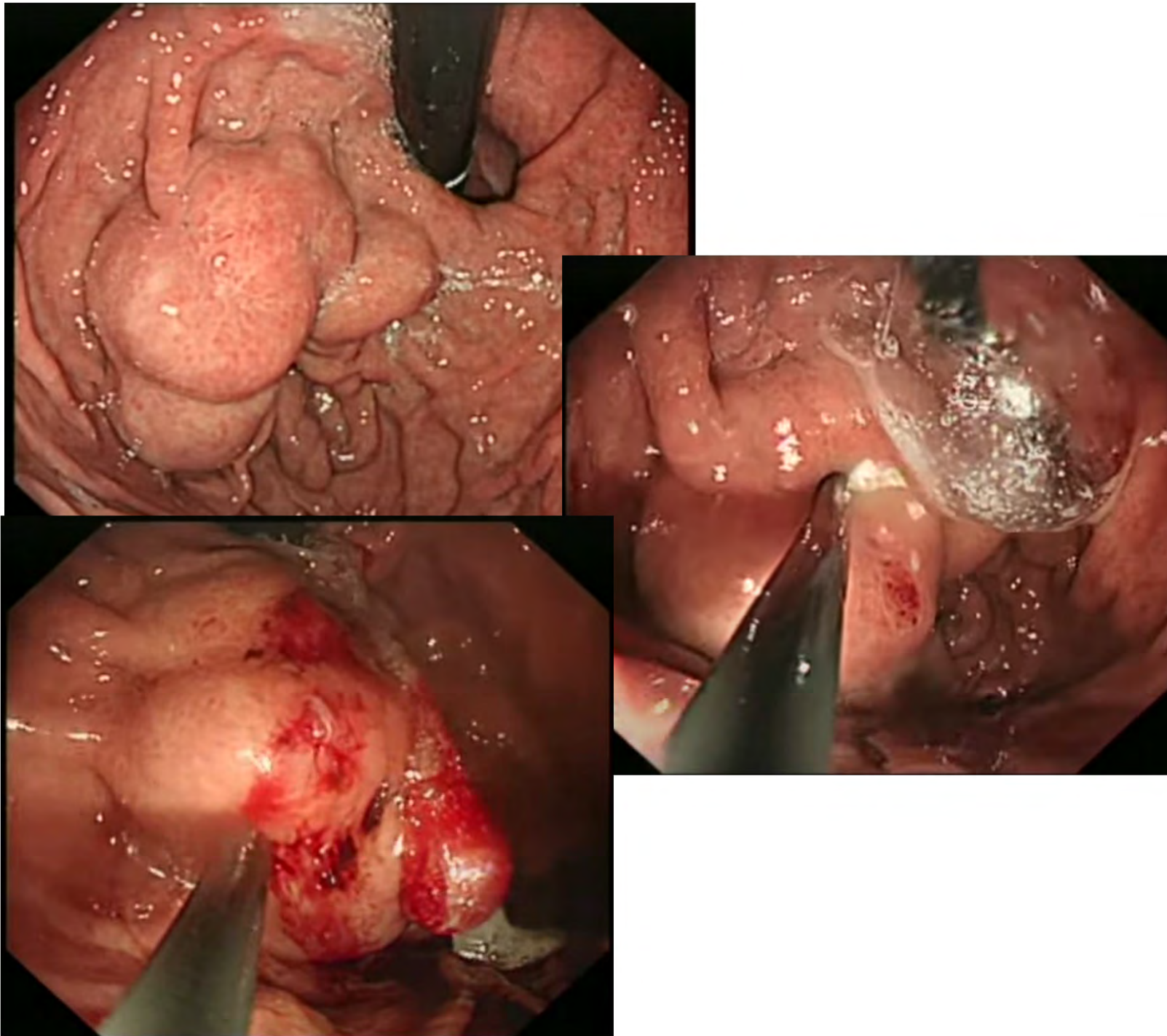
Esophageal stent
(still experimental)



Shunt Surgery
(seldom needed)



Fundal / Gastric Varices

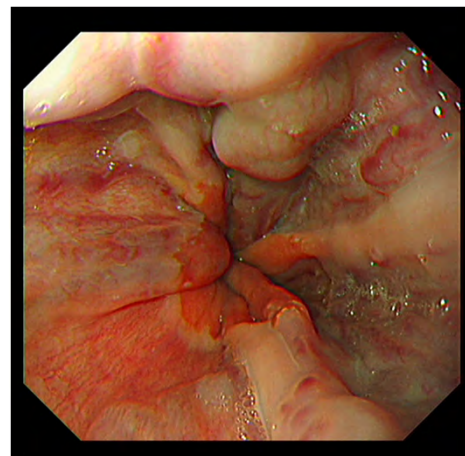


- Cyanoacrylate (tissue glue) injection
- Other management similar to esophageal variceal bleeding

Screening & Prevention

- **Screening OGD**

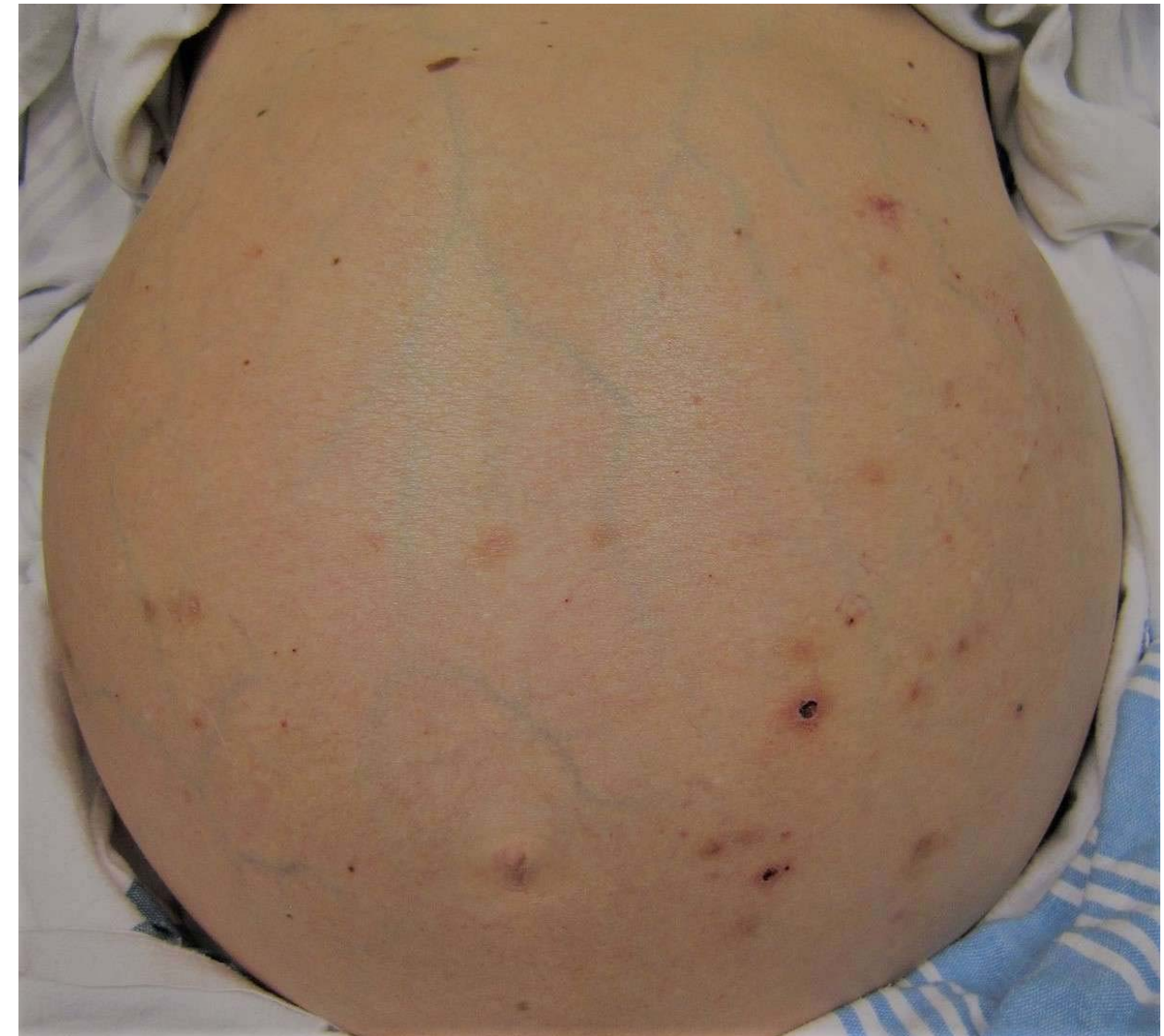
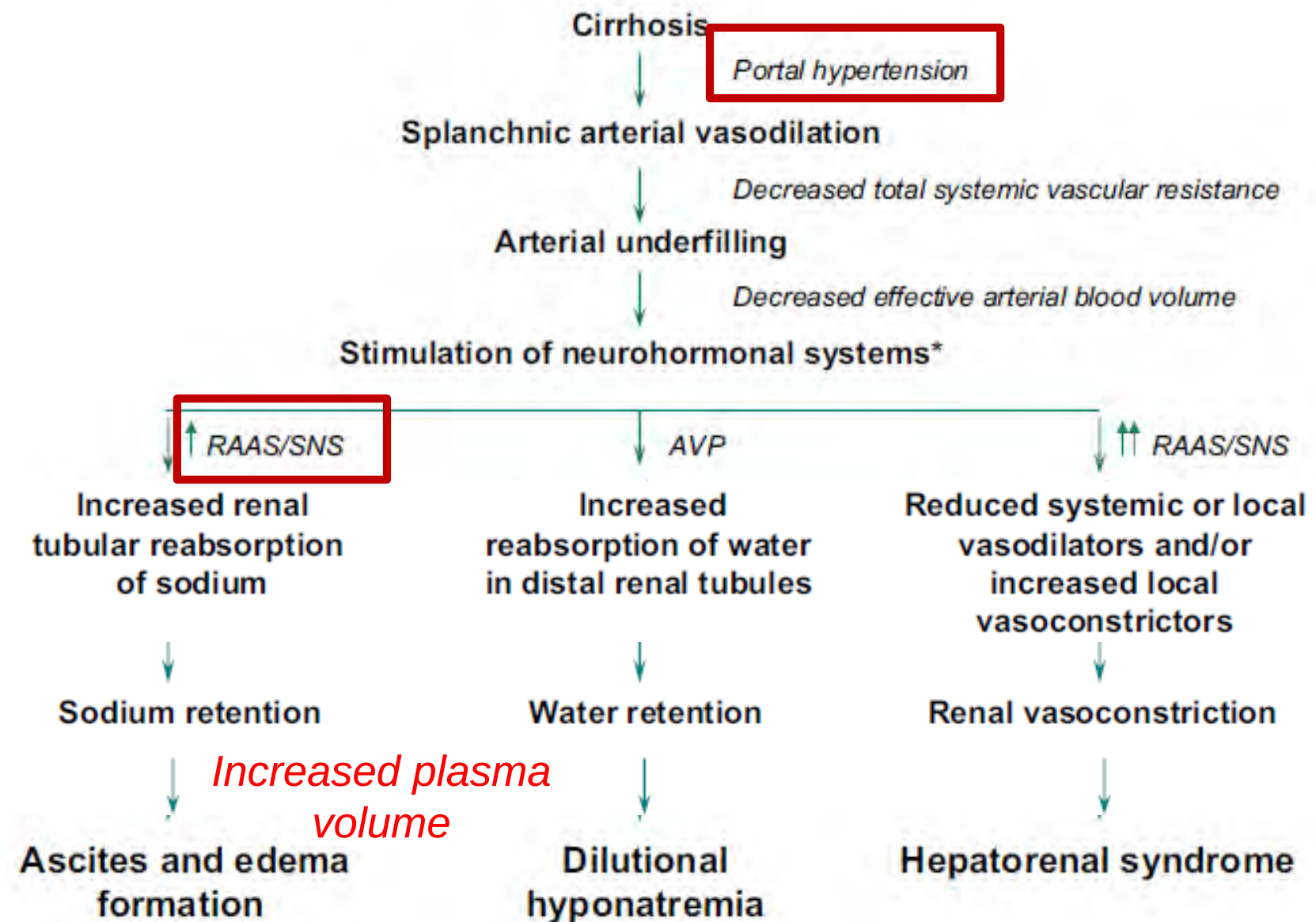
- Cirrhosis
- Not needed if:
 - Liver stiffness ≤ 15 kPa & platelet count $> 150 \times 10^9/L$
(no clinically significant portal hypertension)
- Banding if high-risk lesions present (red-wale signs)



- **Prevention**

- **Non-selective beta-blocker**
 - Oral **carvedilol** / propranolol / nadolol
 - Side effects: hypotension / bradycardia
- Primary & Recurrence prevention
- Survival benefit

Ascites - Pathogenesis



AVP – arginine vasopressin
RAAS – renin-angiotensin-aldosterone system
SNS – sympathetic nervous system

Ascites

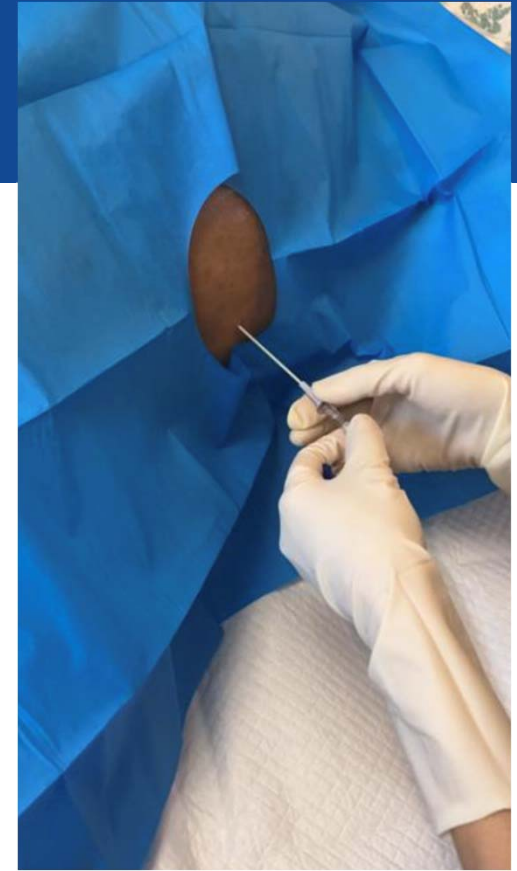
Physical properties

- Colour
 - straw
 - blood-stained
 - chylous
- Protein - may be as high as 40 g/L without special significance
- WCC < 500/mm³ (< 250 PMN)
- Malignant cells < 10% positive in malignant ascites



Diagnostic Abdominal Paracentesis

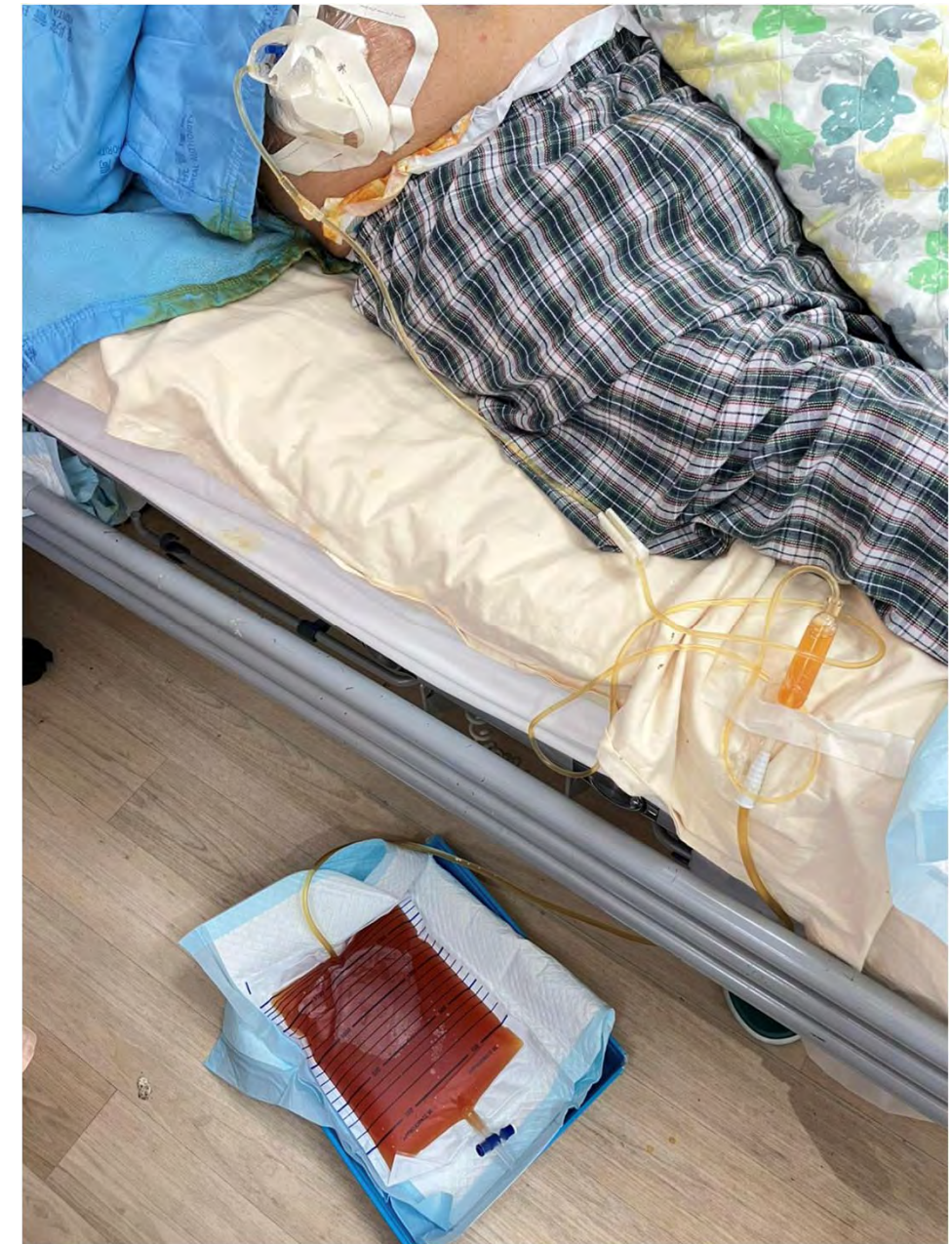
- Generally indicated for new-onset ascites
- Tests include:
 - White cell + neutrophil (PMN) count
 - Total protein
 - Albumin for Serum-ascites albumin gradient (SAAG)
 - Portal hypertension > 11 g/L
 - Other causes of ascites e.g. TB peritonitis, nephrotic syndrome, pancreatitis < 11 g/L
 - Gram stain + culture
 - Cytology
 - Others (amylase, triglyceride, ADA or adenosine deaminase)



Management of Ascites

- Low salt diet ($\frac{1}{2}$ - 2 gms/day)
- Fluid restriction (<1L / day)
- Diuretics
 - Potassium-sparing diuretics +/- Loop diuretics
 - Spironolactone*
 - Amiloride
 - Eplerenone
 - Furosemide
 - Bumetadine
- Aware of overdiuresis / acute kidney injury

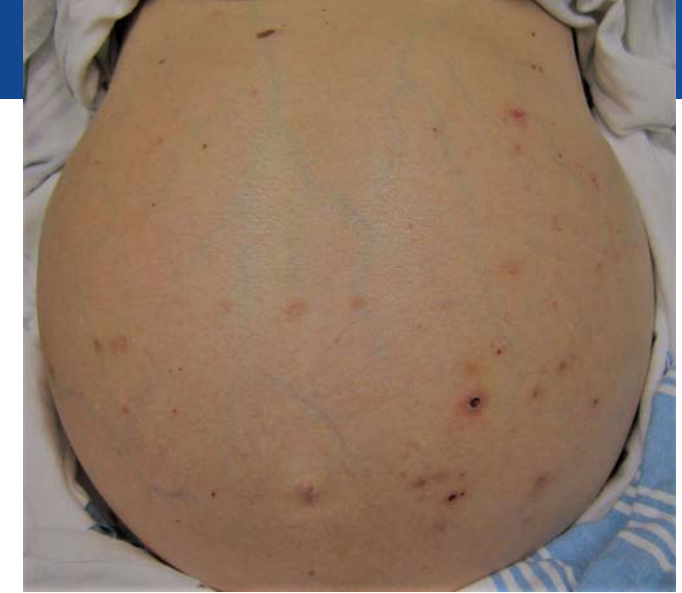
DO NOT give loop diuretics alone



*Gynaecomastia

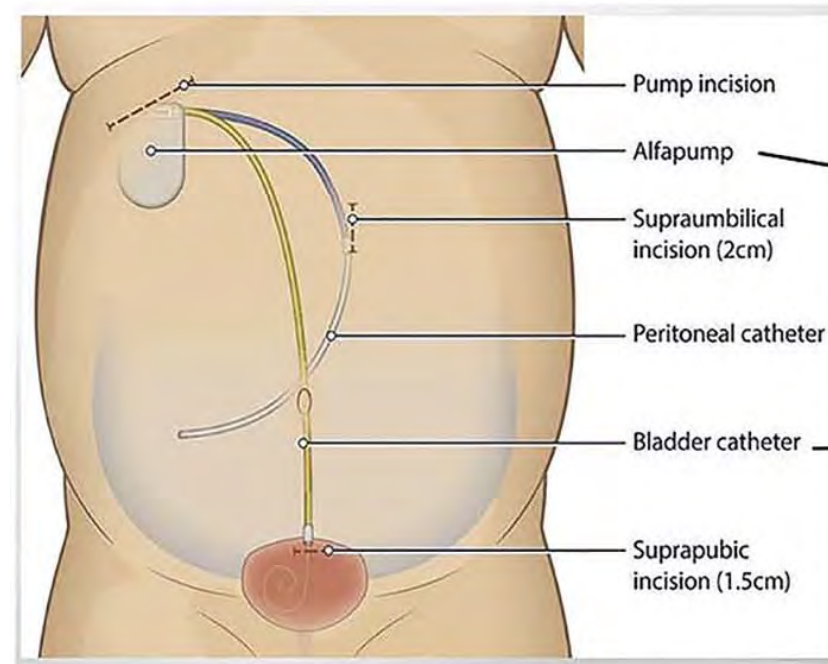
Management of Diuretic Refractory Ascites

- Therapeutic paracentesis
 - For **diuretic-refractory TENSE** ascites
 - 4 – 5 L (up to 8 L)
 - iv albumin (colloid) replacement, especially if paracentesis >5 L
 - Complications (rare):
 - Acute kidney injury (inadequate albumin replacement)
 - HE
 - Perforation of caecum (R-sided puncture)

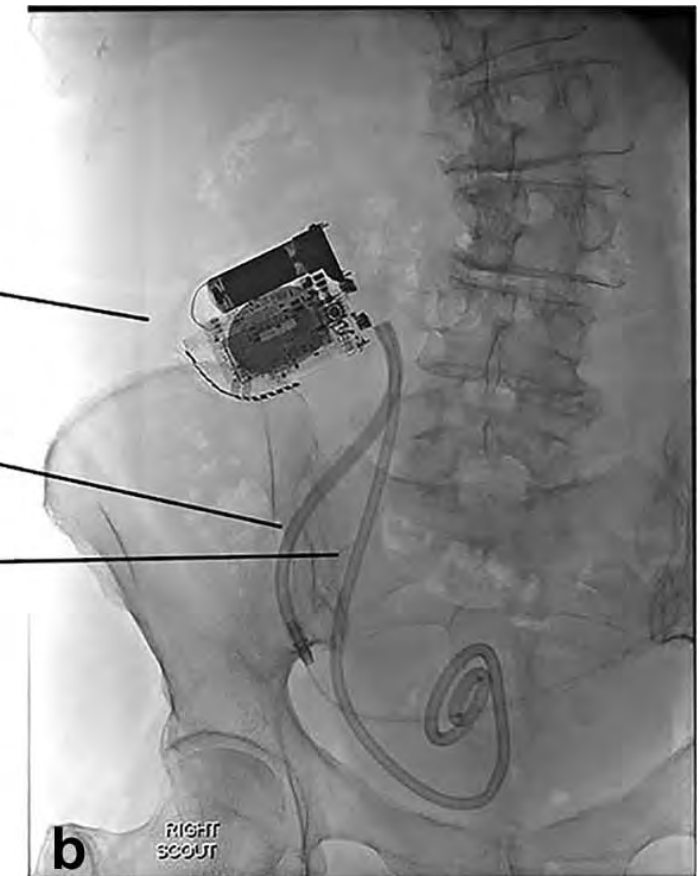


Management of Diuretic Refractory Ascites

- TIPSS
- Experimental
 - Long-term intermittent albumin administration
 - Low-flow ascitic pump

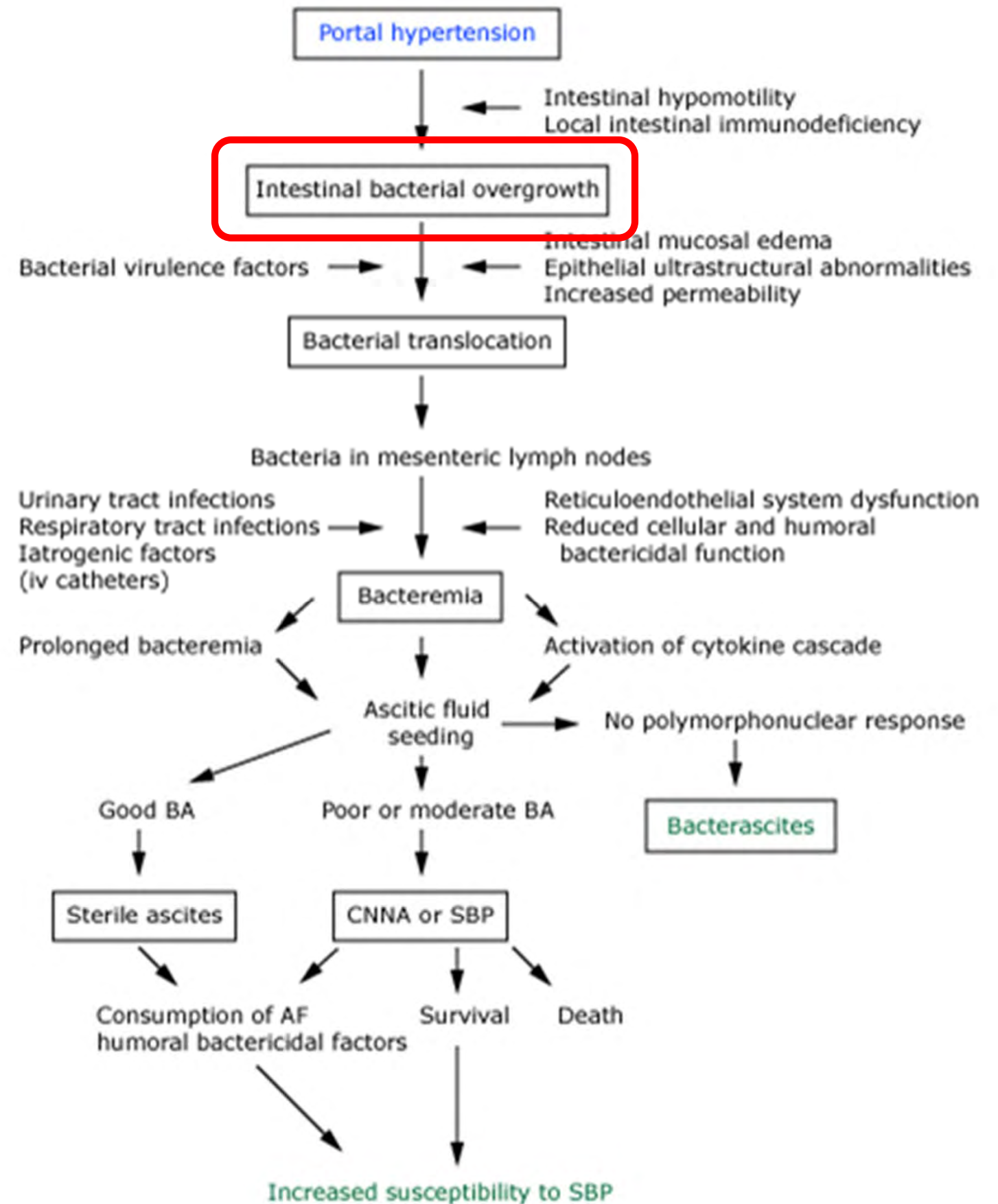


a



b

Spontaneous Bacterial Peritonitis (SBP) Pathogenesis



BA: bactericidal activity; CNNA: culture-negative neutrocytic ascites; SBP: spontaneous bacterial peritonitis.

Spontaneous Bacterial Peritonitis

- Symptoms:
 - Typical: Fever / abdominal pain / new-onset ascites
 - Can have atypical symptoms or asymptomatic
 - Need **HIGH INDEX OF SUSPICION!**
- Mortality: 90% (untreated); <20% (early treatment)
- Diagnosis:
 - **Ascitic PMN count ≥ 250 cell/mm³**
 - Often culture negative

Spotaneous vs Secondary Bacterial Peritonitis

Important Differentials

Other causes of acute abdomen
Bowel perforation

Look out for:

- Surgical signs
- Atypical presentation
- Polymicrobial culture growth
- Lack of treatment response

Spontaneous Bacterial Peritonitis

Treatment & Prophylaxis

- Treatment
 - **Empirical** antibiotics(do not wait!)
 - Intravenous 3rd-generation cephalosporin 5-7 days (cefotaxime / ceftriaxone)
 - Consider carbapenam if severe disease
 - IV albumin (prevent progression to AKI)
- Reassessment diagnostic paracentesis after ~5 days
 - If PMN ↓ can stop antibiotics
 - If PMN no ↓, consider alternative diagnosis
- **Prophylaxis** to prevent recurrence
 - Fluoroquinolones (levofloxacin / ciprofloxacin)

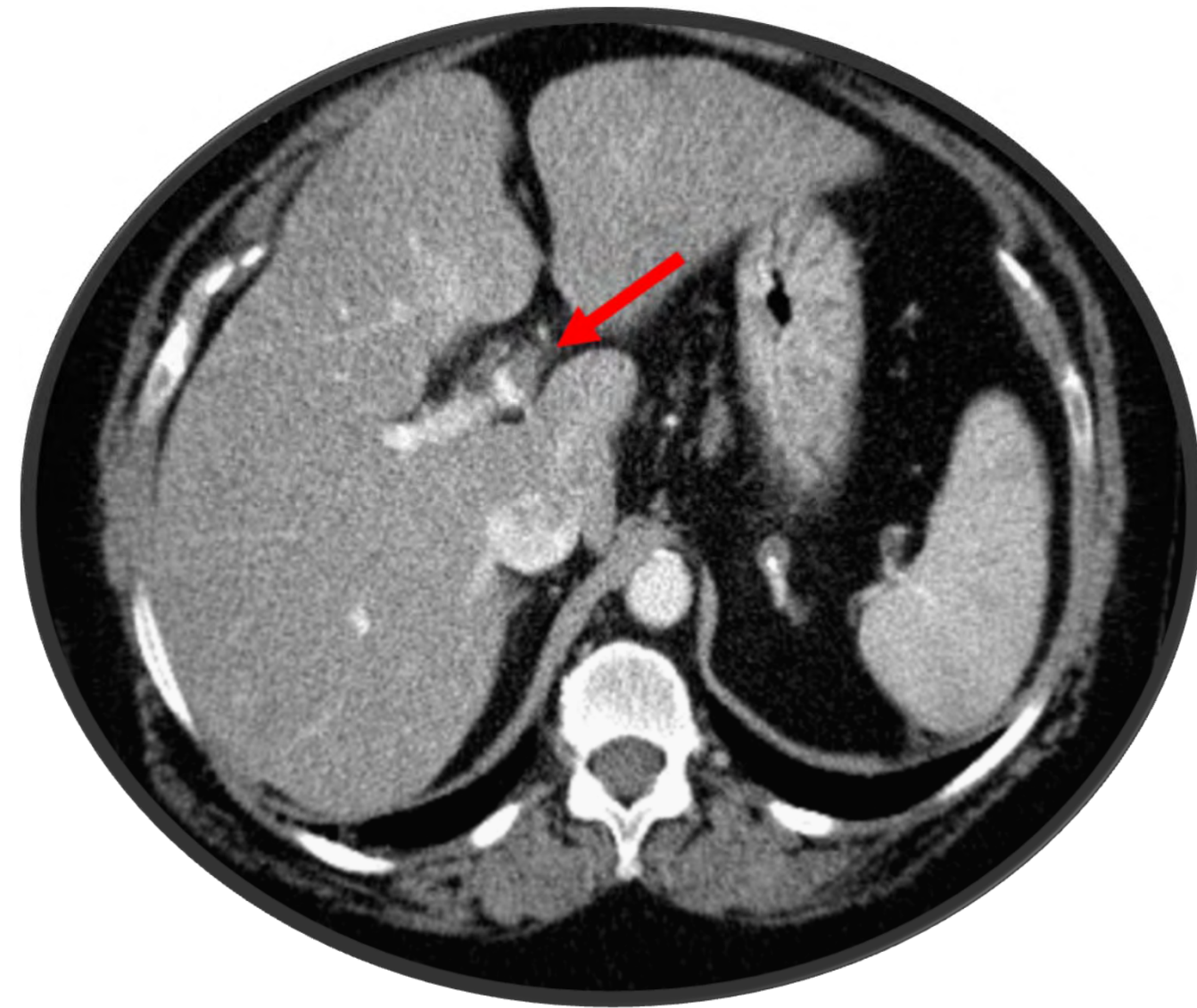
Hepatic Hydrothorax

- Small diaphragmatic defects
 - Peritoneal fluid entering pleura
 - $R > L$
- Negative intrathoracic pressure
 - Pleura fluid $>$ peritoneum
- Transudative pleural fluid biochemistry
- Must **EXCLUDE OTHER CAUSES** of pleural effusion
- Management
 - Ascites management
 - Thoracocentesis / pleuradesis / thoracoscopic repair



Portal Vein Thrombosis (related to cirrhosis)

- Pathogenesis: Slowing of portal vein flow
- >10% of cirrhosis
 - ↑ during decompensation (~25%)
- Symptoms
 - Majority: Clinically silent
 - Minority: abdominal pain when new-onset
- When chronic (cavernous transformation):
 - ↑ variceal bleeding
 - ↑ ascites
 - ↓ treatment options if develop HCC



Portal Vein Thrombosis (related to cirrhosis)

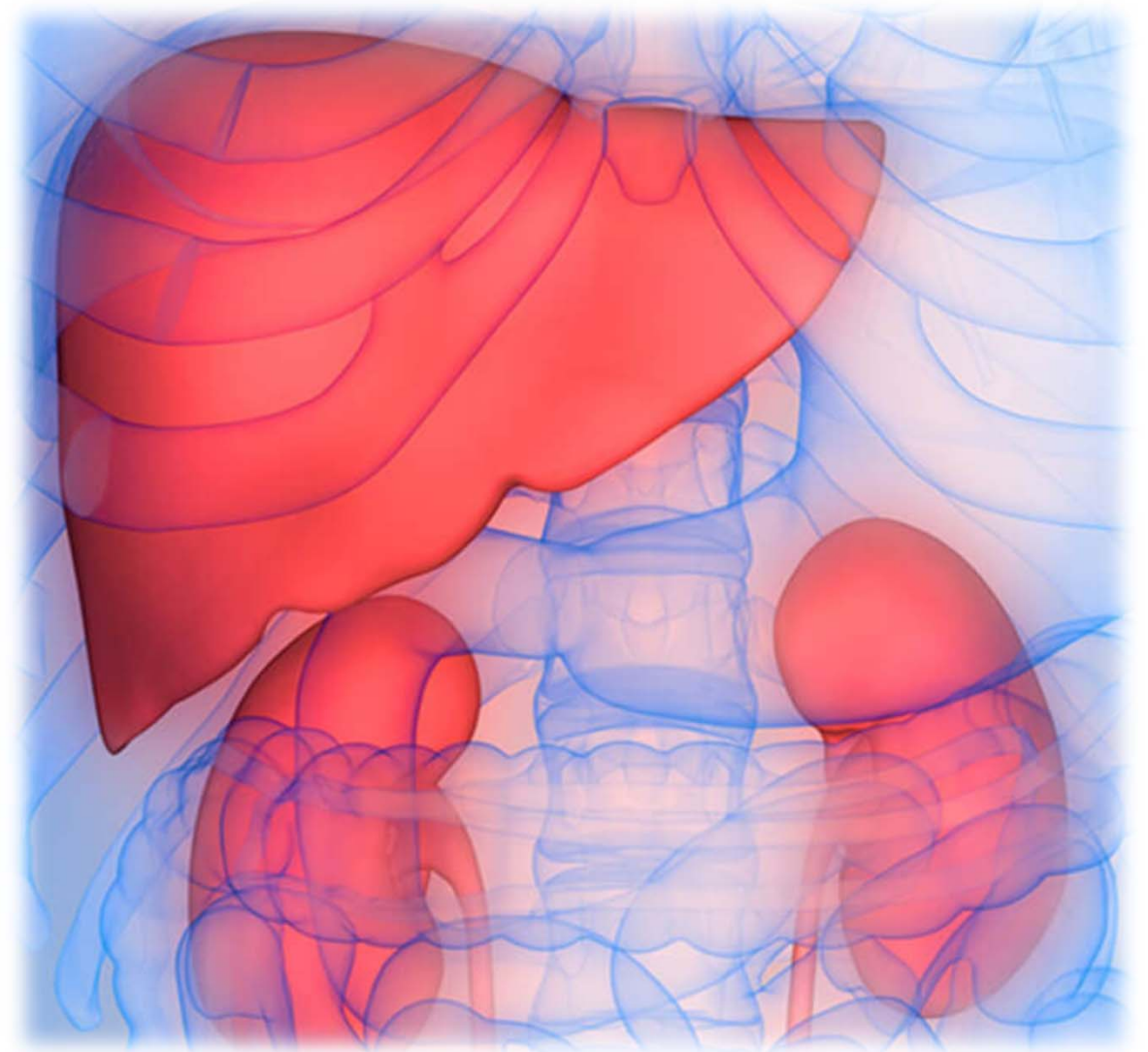
- Diagnosis: ultrasound doppler or CT (contrast)
- Consider other causes of PV thrombosis
 - Malignancy (e.g. pancreatic, cholangiocarcinoma)
 - Venous thrombosis / hypercoagulability causes
 - Septic (e.g. alcoholic pancreatitis)
- Management:
 - Acute stage: Anticoagulation: Warfarin vs NOAC
 - Chronic (cavernous transformation): Anticoagulation is controversial
 - Balance bleeding risk ($\uparrow$ INR, $\downarrow$ platelet)

Acute Kidney Injury in Liver Disease

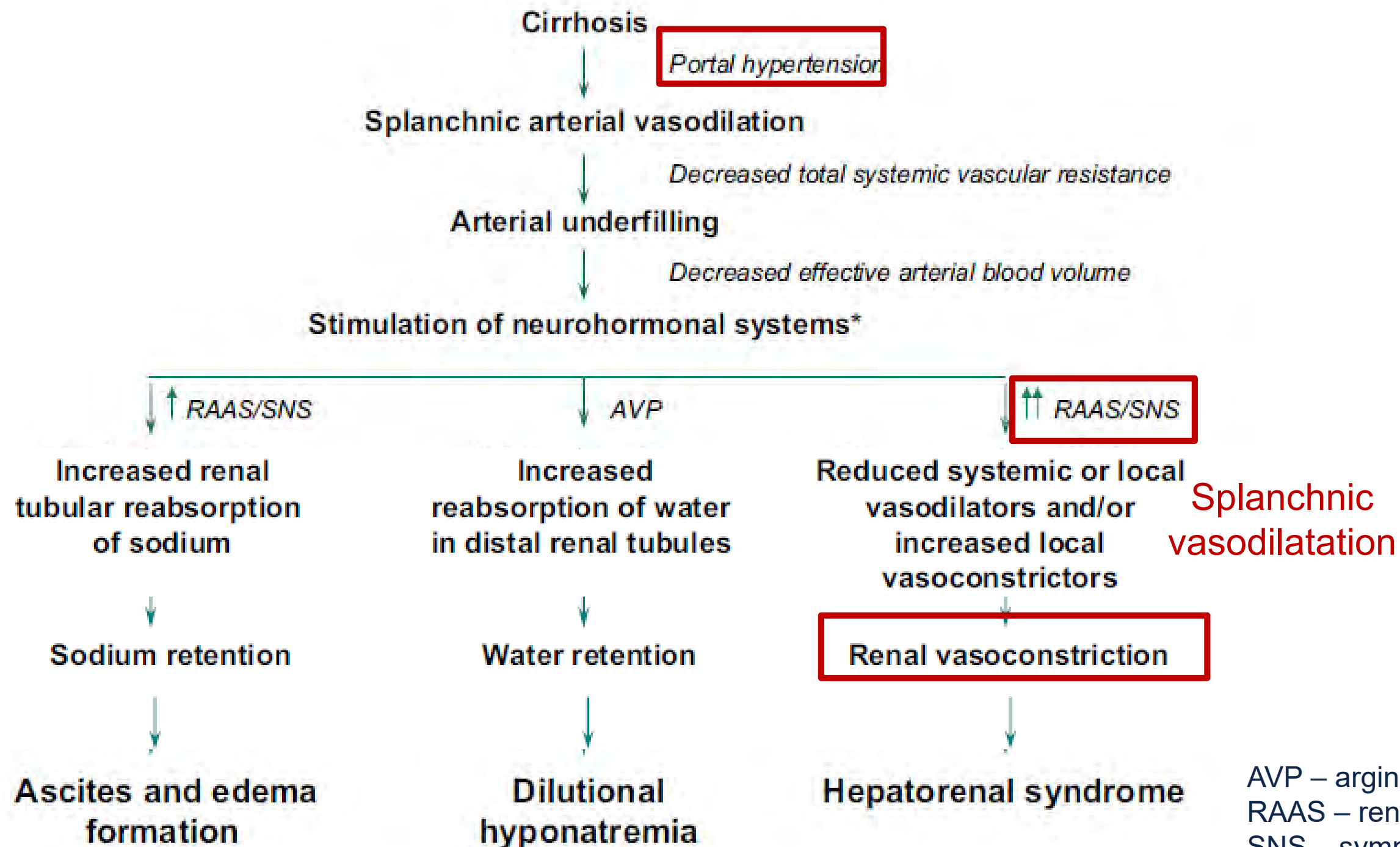
↑ $\geq 50\%$ Cr in 7 days or ≥ 26.5 $\mu\text{mol/L}$ Cr in 48 hours

- Hepatorenal Syndrome (not the most common cause!)
- Hypovolemia – diuretics, GI bleeding
- Sepsis
- Nephrotoxic drugs / herbs
- Parenchymal renal disease
- Obstructive uropathy

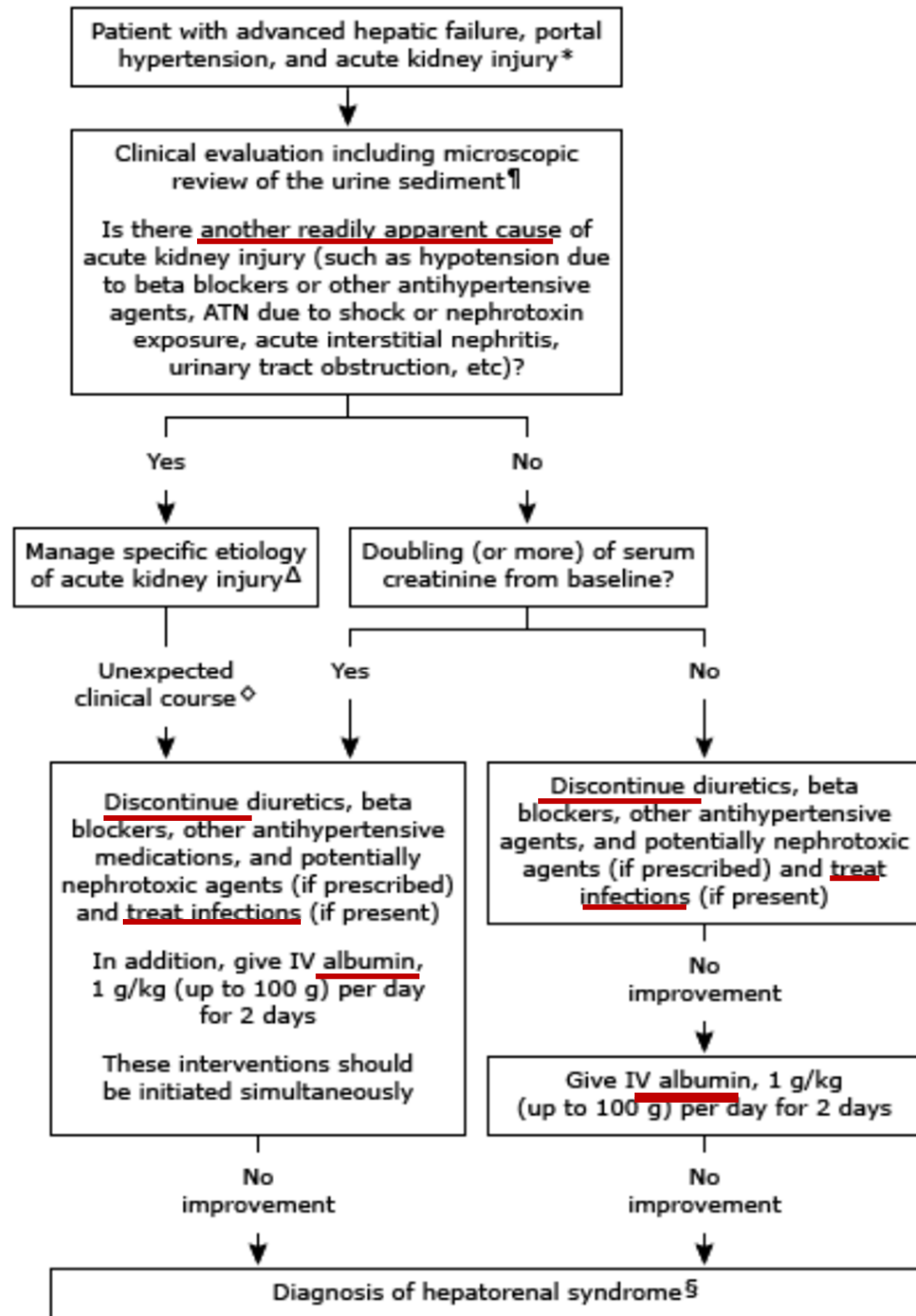
Always exclude other causes of AKI!



Hepatorenal Syndrome: Pathogenesis



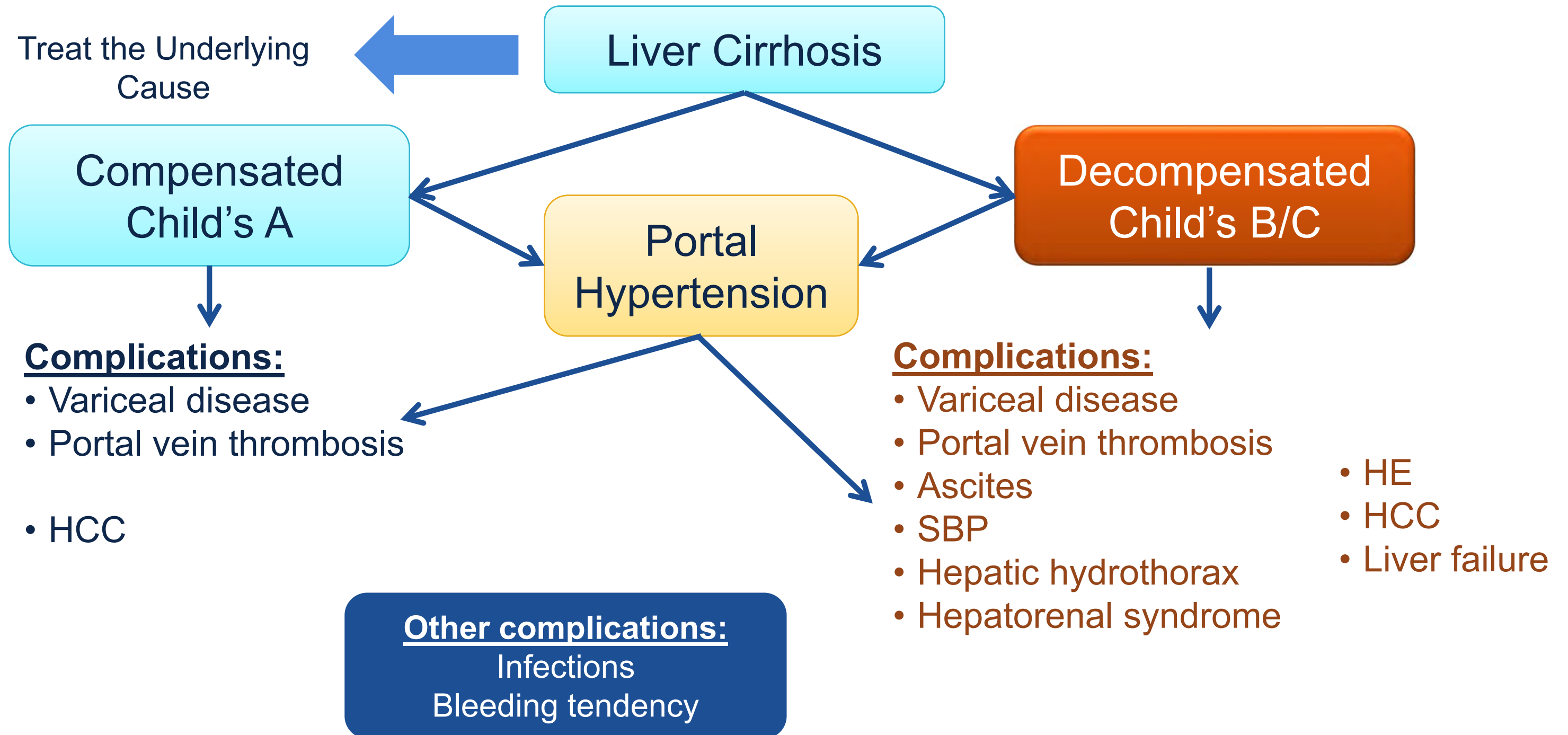
AVP – arginine vasopressin
RAAS – renin-angiotensin-aldosterone system
SNS – sympathetic nervous system



HRS-AKI Management

- Simultaneous management
 - Exclude other causes
 - Discontinue potential culprit drugs
 - Diuretics, NSBB, etc
 - Treat infection
 - IV Albumin
 - IV Terlipressin
 - Spot urine sodium <10 mmol/L suggestive of HRS
 - Can recur after treatment discontinuation
 - RRT / LT...
- ➔ **Terlipressin**

The Cirrhotic Patient: One-Slide Overview



WCS Outline

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- Portal Hypertension: pathogenesis
 - Variceal disease
- Ascites: pathogenesis
 - Ascites
 - Spontaneous bacterial peritonitis
 - Hepatic hydrothorax
 - Portal vein thrombosis
- Acute Kidney Injury / Hepatorenal syndrome

THANK YOU!

*Email questions to
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Clinical Professor

The University of Hong Kong

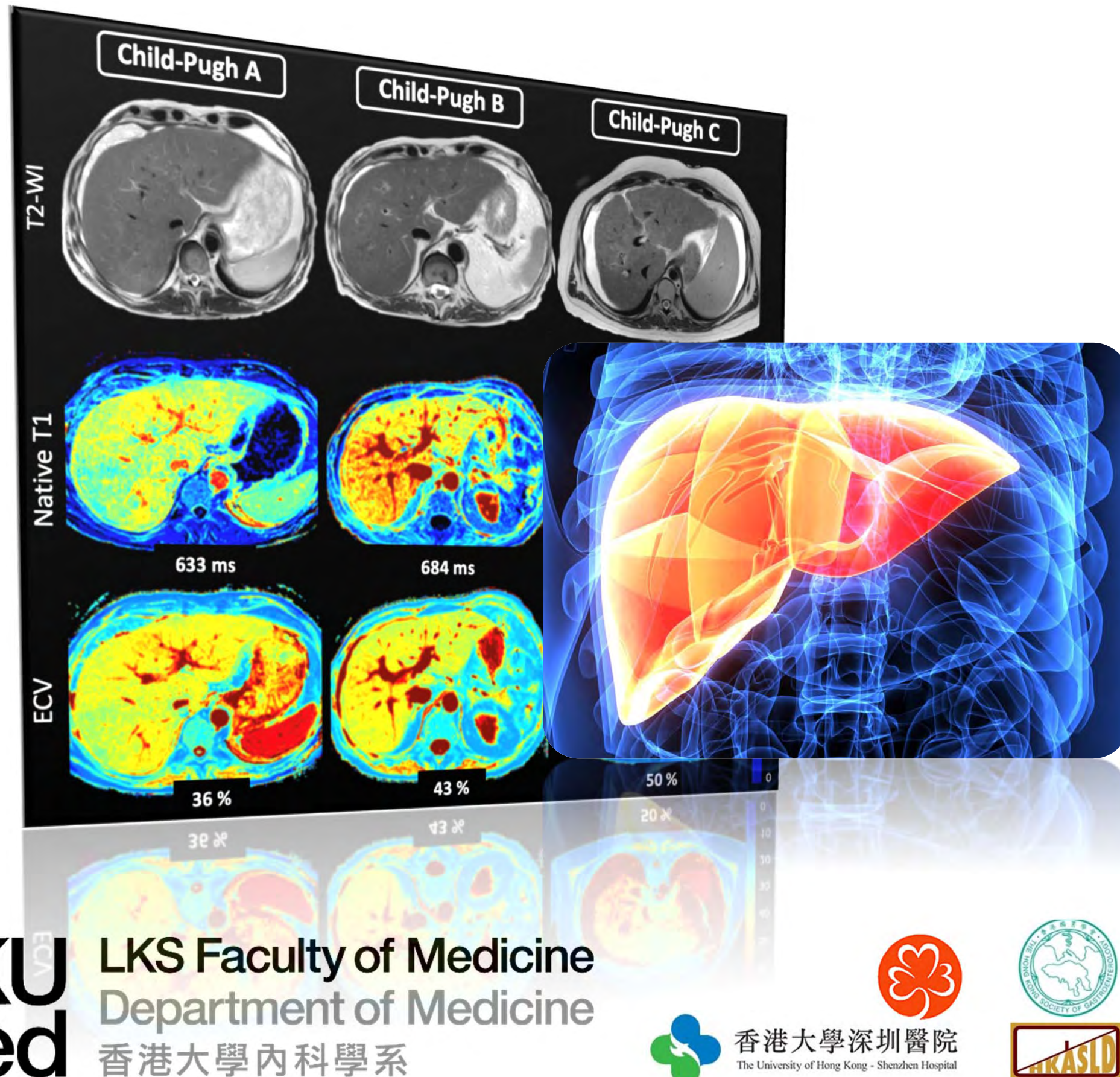


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LKS Faculty of Medicine
Department of Medicine
香港大學內科學系



香港大學深圳醫院
The University of Hong Kong - Shenzhen Hospital



TELL US WHAT

JC Whole Class Session

YOU THINK

THANK

YOU!



Look for today's teacher
and the others you may have missed.
(HKU Portal login required)